Competing switching and phase-escape channels in hafnia: a mode-strain pathway analysis

Abstract

Mechanical constraints can alter polarization-reversal paths while also changing the routes by which a polar crystal leaves its original phase. We examine this competition in twelve-atom HfO2 using first-principles variable-cell nudged elastic band calculations under a common partially clamped boundary. Four ordered candidates connect the same polar initial well to opposite-polar variants or tetragonal and monoclinic structures. The registered switching candidates respond oppositely to +1% substrate strain, despite equal final-well responses. Signed oxygen-shuffle coordinates distinguish their atomic routes, but do not exhaust their accommodation. With the displayed patterns and polar z proxy fixed, atomic release at the zero-strain preserving bottleneck recovers 13.6 meV/HfO2 through mixed motion with a transverse structural component. Reconnecting this configuration and relaxing every interior yields a sampled peak of 83.2 rather than 96.8 meV/HfO2, close to the examined tetragonal escape peak of 84.4 meV/HfO2. A prospectively frozen +0.5% test gives preserving and escape peaks of 98.5 and 90.4 meV/HfO2. Subsequent initialization controls change the preserving peak by 10.8 meV/HfO2 but the escape peak by only 0.13 meV/HfO2. Thus a compact mode projection and an ordinary force pass alone do not make the inferred competition or strain nonlinearity initialization-independent. The measured local intervention and lower connected candidate link this limitation to atomic accommodation beyond the displayed patterns. The conclusions concern finite sampled paths under the specified boundary, not certified saddles or a global ordering; the independent tests establish no additional predictive advantage of the local reduced models.

1. Introduction

Ferroelectric hafnia combines polar and nonpolar structures that can be connected by relatively small atomic shuffles and lattice deformation. A useful polar state must support reversal while resisting transformation into another structural phase. These requirements involve different trajectories on the energy landscape: reducing a switching barrier can also change the barriers for leaving the polar well. The relevant mechanical-control question is therefore how an imposed constraint changes the competition between these routes from one common initial state. This distinction matters even before effects such as domain-wall motion, defects and temperature are introduced.

Prior work establishes both the mechanical sensitivity of hafnia pathways and their connection to lattice distortions. Liu and Hanrahan showed that orientation, epitaxial strain and clamping affect phase stability and variable-cell transformation barriers. Behara and Van der Ven described structural variants and their pathways in strain coordinates and connected intermediate shuffle patterns to mechanically constrained domain boundaries. (Liu and Hanrahan, 2019; Behara and Van der Ven, 2022) Mode-based studies identify coupled polar and nonpolar distortions: Delodovici, Barone and Picozzi analyzed trilinear coupling, while Qi, Singh and Rabe showed that reversal can involve distinct accompanying antipolar changes and switching paths. (Delodovici et al., 2021; Qi et al., 2025) Thus neither the existence of coupled modes nor the multiplicity of switching routes is the new claim of this work.

Reference choice and boundary conditions further shape the physical interpretation. A Cmma reference organizes competing phases through unstable phonon bands, while strain-dependent mode models describe changes in the tetragonal-to-polar landscape. (Qi and Rabe, 2025; Lee, Lee and Yu, 2026) Constrained-mode landscapes and subsequent NEB checks have also been used to study an antipolar switching intermediate. (Zhou et al., 2022) The energetic role of Pbcn depends on the functional and

mechanical ensemble; fixing a PO-referenced lattice gives a different comparison from allowing selected cell directions to relax. (Fan et al., 2025) Likewise, a phase-pair name alone does not specify the ordered atomic mapping, as illustrated by distinct M-to-PO candidates in the public Ma–Liu study. (Ma and Liu, 2023, v3 supplement) Phonon-pair condensation can also explain low-energy domain walls, although an interface energy and a homogeneous transformation profile describe different physical processes. (Lee et al., 2026)

These results motivate a comparison that keeps three questions together. Which examined route has the lower peak from the same polar well? Which atomic rearrangements and permitted strains distinguish the routes? How far does their observed response transfer to an independently evaluated mechanical condition? Endpoint energy shifts answer only part of the first question, and a compact displacement projection does not by itself answer the third. A useful analysis must connect the full path energies to actual structures and then test the explanatory or predictive role of selected coordinates.

Here we compare two ordered reversal candidates and escape to tetragonal and monoclinic HfO2 under one T-referenced substrate constraint. We use the polar orthorhombic Pca21 structure, denoted PO, as the common initial well; T and M denote the tetragonal P42/nmc and monoclinic P21/c structures. The reversal candidates preserve or reverse a signed reference oxygen shuffle while reversing the polar structural displacement. VARNEB provides the common atomic-cell path representation and energy/force/stress interface. We combine their energy profiles with signed shuffle amplitudes, open-cell deformation and matched frozen, atomic-only and joint relaxation controls. Finally, forecasts fixed before the target-interior calculations are compared with two +0.5% paths. The contribution is a common-reference analysis linking channel competition to atomic and lattice accommodation, testing a local relaxation explanation and checking response transfer at an independently evaluated strain. Keeping the candidate identities, permitted deformations and energy reference together makes the physical comparison explicit.

2. Theory and computational approach

2.1. Joint pathways and a common energy reference

Ordinary NEB relaxes the physical force perpendicular to the path while springs control image spacing. (Henkelman and Jonsson, 2000) Atomic-cell extensions incorporate lattice degrees of freedom with an explicit relative metric. (Sheppard et al., 2012; Qian et al., 2013) VARNEB applies these established path ideas through energy, force and stress results obtained via ASE-compatible calculators. (Larsen et al., 2017) The backend, optimizer and mechanical active space are independent choices. Ordered atoms, reference cells and continuous periodic lifts are preserved during optimization and analysis; an image-wise remapping is not a physical pathway improvement.

For candidate channel α at condition c, define its bottleneck barrier from the same initial well I as

$$B_\alpha(c) = E_\alpha^\dagger(c) - E_I(c).$$

Here a stationary bottleneck and a discrete sampled maximum are distinguished; the latter is denoted by a hat in the pilot interpretation. With a common I, the initial-well energy cancels from $B_\alpha - B_\beta$. A change in relative channel selectivity thus cannot be explained by shifting that common well alone. In the declared candidate set, we use

$$B_s = \min_{\alpha\in\mathcal{S}} B_\alpha, \qquad B_d = \min_{\alpha\in\mathcal{D}} B_\alpha, \qquad S = B_d - B_s.$$

An increased S is relative selectivity, not improved absolute resistance to escape. The stronger prospective test requires a resolved reduction of B_s and a lower uncertainty bound on ΔB_d no smaller than zero. An unresolved decay response is inconclusive, not evidence of noninferiority. This primary allowable-loss

margin is fixed before boundary-response labels. Both T and M escape candidates are retained. These are candidate-set statements, not an exhaustive network or a calculation of device lifetime. At finite prescribed pressure the objective would be $E + PV$; all hafnia results here instead use $P = 0$ and E.

2.2. Mechanical boundary, ordered coordinates and mode interpretation

The partially clamped ensemble fixes two T-referenced in-plane cell vectors and releases the third vector, including its two permitted tilt components. Every endpoint and image shares that allowed space. In the original T chart, x and z follow the prescribed substrate strain, while y remains open. Zero substrate strain denotes the unscaled reference plane and differs from a fully stress-free crystal. At each condition the initial and final wells are separately relaxed under the same boundary.

We preserve atom order and a continuous periodic lift over each chain. For the structural interpretation in Figure 3, fractional displacements are expressed in one original-T Cartesian metric after removal of rigid translation. Three orthonormal geometric oxygen-shuffle patterns, obtained from the tetragonal distortion and its rotated variants, define signed amplitudes Q_x, Q_y and Q_z in Angstrom. The patterns describe the path geometry and do not assign phonon irreducible representations or energies to individual modes. A polar-displacement descriptor is the mean O minus mean Hf displacement along the original polar z direction; electronic polarization requires a separate calculation.

Cell response is reported separately through the deformation gradient F. The plotted open-axis stretch is $100(F_{yy} - 1)$, distinct from the Green-strain component and from an elastic energy. Atomic projections, cell stretch and their joint energy response answer different questions. The alternative mass-weighted T optical and Cmma reference comparisons remain in Appendices D and F, with every original rank and registration retained. Coordinate and boundary identities are not selected from the independent labels.

For the free-cell/clamped comparison, each band has its own relaxed endpoints and initial energy. A constraint orders ideal minimax barriers only when those endpoints and the thermodynamic potential are common. The present relaxed-endpoint comparison does not meet that premise. Appendix B gives a simple counterexample; Appendix G retains the detailed boundary and response derivations.

2.3. Electronic calculations and matched local relaxation

The production model is periodic Hf4O8, twelve atoms and four formula units, at zero temperature, external pressure and electric field. Calculations use ABACUS/PBE, the original Hf/O pseudopotentials, full 10-au DZP orbitals generated at 100 Ry, ecutwfc=100 Ry and the original Gamma-centred 2x2x2 mesh. The same input, pseudopotential and orbital bytes apply to every comparison. The unchanged electronic settings are scf*thr=1e-8, scf* nmax=150 and mixing_beta=0.3. (Li et al., 2016; Chen et al., 2010; Perdew et al., 1996) Byte-exact INPUT and KPT files accompany the data. A calculation at a changed geometry always obtains new energy, forces and stress; reuse is restricted to byte-linked static results at exactly identical geometries and electronic settings.

The clamped channel comparisons use nine total images, including seven moving interior images, and the unchanged ordinary NEB residual target 0.10 eV/Angstrom. No climbing image is used. The ten-image historical T-to-PO formation band, finite sampling supplements and free-cell candidates are retained separately in Appendix C. These are distinct calculations rather than alternative resolutions silently pooled into one optimized path. A residual pass determines ordinary convergence for the declared polygon and does not bound its continuous-barrier error.

To distinguish local atomic accommodation from additional cell response, we compare three treatments at fixed retained joint coordinates $q = (Q_1, Q_2)$: a frozen configuration F, atomic-only release A at that configuration's cell, and joint atomic/open-cell release J. Each of the preserving and T sampled-peak neighborhoods uses its origin and four axial q pairs. The local chart has 33 translation-free atomic and three open-cell length coordinates with scale $L = 5.12968067458423$ Angstrom; q is not the geometric

Q_x/Q_z projection. Retained coordinates are held fixed. The atomic-only treatment releases 31 atomic directions; joint release also allows the permitted cell response.

To connect these local controls to the path descriptors, we reconstruct all F/A/J geometries in the same ordered reference chart and project their translation-free atomic displacement differences onto the three geometric T patterns of Section 2.2 and the complete orthogonal complement. Norms use the original T Cartesian metric over all twelve atoms, without mass weighting; their squared fractions describe geometry, not energy.

The pointwise differences $E_A - E_F$ and $E_J - E_A$ measure the observed treatment contrasts. The latter includes atomic readjustment as well as cell motion, so it is not a unique strain-energy term. The released-gradient target is 0.03 eV/Angstrom, distinct from the NEB threshold. Finite-step curvature probes, branch trials, model definitions, geometry guards and actual costs are retained in Appendices D–G. These diagnostics do not replace the measured path peaks or imply stable conditional minima.

A separate across-strain control retains the four physical observables Q_x, Q_z, Q_y and the polar displacement proxy along original z, rather than the local joint coordinates Q_1, Q_2. Their definitions are common to the three PO-to-T sampled peaks at 0, +0.5 and +1% strain, but their native values differ. Each F/A/J triplet holds its own four values exactly. After fixing three translations, A releases 29 atomic directions at exactly F's cell; J starts from A and adds the three open-cell directions, giving 32 released coordinates. The same 0.03 eV/Angstrom gradient target and electronic contract apply. A bounded quasi-Newton initialization repair retained the original unsuccessful attempts and reused successful native results exactly; it did not modify the physical calculation or replace the whole-path labels. These four-observable controls are not nested with the earlier two-coordinate controls.

For the preserving neighborhoods, the same four-observable restriction was tested from two signed 0.10 Angstrom atomic starts along the omitted negative direction of the existing local curvature calculation. Each A or J arm allowed at most twelve fresh attempts, with a 0.5 Angstrom full-coordinate neighborhood bound and no electronic-parameter changes. Only a successful A candidate could seed J and two 0.01 Angstrom probes along that known direction. These are finite branch tests, not an exhaustive search or a full post-release stability calculation.

One zero-strain negative-start J candidate was then inserted as the free central image of the original nine-image band. Both endpoints and their native static caches were retained, while all seven interiors were free; the four local observable constraints were not carried into NEB. This connection test allowed at most 24 FIRE updates and used the same ordinary 0.10 eV/Angstrom target, spring and step limits, with no CI. It tests a complete connected candidate rather than subtracting a local relaxation energy from the original barrier.

2.4. Prospective strain test and prediction comparison

Eight training bands compare all four candidates at substrate strain 0 and +1%. Before generating target structures, the lower sampled switching and escape representatives at zero strain were selected: preserving reversal and PO-to-T. Their ordering at a new condition was not used to change that selection. Three +0.5% endpoints were then optimized under the imposed plane with atomic force at most 0.03 eV/Angstrom and open traction below 2 kbar. Endpoint features were shared equally by the forecast arms that require them. The complete numerical and unavailable prediction panel was frozen before evaluating either target path's interiors.

B0 predicts the arithmetic midpoint of each zero/+1% sampled peak. With zero-strain anchor $\widehat{B}(0)$ and measured initial/final well shifts $\Delta E_I, \Delta E_F$ per formula unit, the two endpoint controls are

$$\widehat{B}_{\text{B1,fixed}} = \widehat{B}(0) - \Delta E_I, \qquad \widehat{B}_{\text{B1,follow}} = \widehat{B}(0) + \Delta E_F - \Delta E_I.$$

All local response models and reference variants remain in the panel. A model that had not supplied a valid transfer to the physical strain control records an unavailable prediction rather than an inferred value. Errors retain their signs; channel errors and the T-minus-preserving error use the same initial-well convention.

Each target band starts from its corresponding zero-strain polygon, with the plane affinely changed and the new endpoint corrections distributed along image index. The fixed recipe uses FIRE, $k = 0.2$ eV/Angstrom squared, maxstep 0.02 Angstrom, maximum cell step 0.02, minimum pair distance 1.6 Angstrom and maximum deformation 0.25. It permits at most 80 updates, four candidate-step retries and 24 hours per path. The nine-image target and 0.10 eV/Angstrom criterion are unchanged. Both paths use native 32-rank mpirun allocations with one thread per rank.

The scored observable is the endpoint-inclusive sampled peak relative to the common PO+ well. We report its change from zero strain, the selected-pair difference, every frozen error or abstention, and actual SCF/allocation cost. The complete preregistration and data record the chronology. This fixed-protocol check tests response transfer; it does not search for new channels or resolve continuous-path curvature.

3. Results: competing responses under one mechanical intervention

3.1. Strain changes channel competition, not just one switching peak

All eight registered clamped training bands passed the unchanged ordinary 0.10 eV/Angstrom residual criterion. Each substrate condition used one identical ordered PO+ starting cache for all four candidates; energies at different conditions were referred to their own relaxed PO+ wells. Figure 1 and Table 1 therefore compare the same four registered correspondences, not selected snapshots with different initial-state offsets. The full forward/reverse values and final-well differences are retained in Appendix D, Table A4.

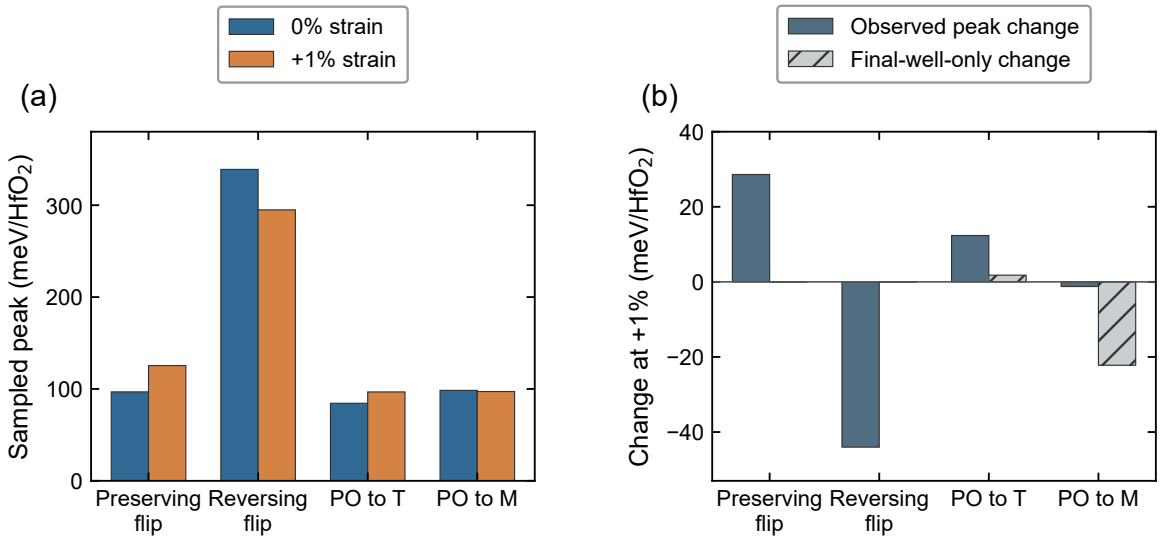


Figure 1. Channel-dependent response under the same partially clamped boundary. (a) Sampled forward peaks at zero and +1% substrate strain, each relative to its condition's common PO+ well. (b) Observed changes compared with the final-well-only B1 response, which assumes that the absolute peak follows the final-well shift. The two switching endpoints have the same well-energy response but their sampled peaks change in opposite directions. All four registered candidates are shown. Bars are deterministic path observations, not replicate means; no barrier-error interval is inferred from the ordinary NEB residual.

Table 1. Sampled forward peaks under the matched boundary, in meV/HfO2. The last column is the +1% value minus the zero-strain value, not a derivative.

Candidate	Zero strain	+1% strain	Change
PO-to-M	98.458122	97.227719	-1.230403
PO-to-T	84.416624	96.763650	12.347027
Preserving flip	96.828183	125.424431	28.596248
Reversing flip	338.968965	294.944634	-44.024331

The preserving and reversing switching peaks changed by opposite signs. The decrease of the reversing peak by 44.024331 meV/HfO2 did not make it the lower examined switching candidate: it remained above the preserving peak at both training conditions. The lower sampled switching peak increased by 28.596248, whereas the lower sampled escape peak increased by 12.347027. The sampled selectivity proxy, escape minus switching peak, consequently changed from -12.411559 to -28.660781 meV/HfO2. Its decrease of 16.249222 is not the favorable signature of simultaneously easing switching and retaining resistance to escape.

This is an observed candidate-set contrast, not an error-certified rejection of the original favorable-response hypothesis or a global pathway ordering. The +1% T/M escape peaks differ by only 0.464068 meV/HfO2, and ordinary residuals do not bound sampled-barrier error or certify stationary bottlenecks. Both escape candidates and both ordered switching candidates remain in the analysis; the decreasing but high reversing candidate is not used alone to claim improved switching. The later zero-strain connection test in Section 3.3 finds a lower preserving candidate, so the registered zero-strain ordering is not robust to the admitted atomic accommodation. Table 1 remains the original training set, not a retrospectively selected set of lowest outcomes.

The response difference is not an initial-energy-offset effect. The common PO+ well rises by 6.806502 meV/HfO2. Before subtracting it, the preserving sampled peak rises by 35.402751, whereas the reversing peak falls by 37.217829 meV/HfO2. Thus the opposing switching responses persist on the absolute energy scale. The configurations in the path interior, not merely the shared starting or final wells, must enter an explanation.

Final-well response alone also failed to reproduce these sampled changes. Table 2 gives the preregistered final-following B1 null increments. The alternative fixed-absolute-bottleneck null gives -6.806502 meV/HfO2 for every channel because their initial well is common. For PO-to-T the final-following increment is +1.785908, compared with the observed +12.347027; for PO-to-M it is -22.206479, compared with -1.230403. These are seen training comparisons, not independent predictive wins or error-resolved null rejection.

Table 2. Seen final-well shifts and final-following B1 barrier-response increments, in meV/HfO2. Neither column is a measured bottleneck slope.

Final endpoint representation	Observed well-energy shift	B1 final-following barrier increment
T descendant	+8.592411	+1.785908
M	-15.399977	-22.206479
PO-, T pattern preserving	+6.841761	+0.035259
PO-, T pattern reversing	+6.841761	+0.035259

The complete native training replay retains 72 frame records and 66 distinct raw-log/geometry identities, including the repeated common initial caches. They are not independent statistical replicates. The independent +0.5% test is reported separately in Section 3.4; these training values alone do not establish independent predictive accuracy.

3.2. Clamping changes the sampled energetic role without changing the group label

The preserving paths offer a direct check of the earlier low-energy-centre caveat (Appendix C.2). Reanalysis of the original free-cell step-69 band and the zero-strain clamped terminal band identifies both central samples as Pbcn at each fixed 0.001/0.01/0.05 Angstrom tolerance and 1-degree angular tolerance. No structure is standardized, remapped or relaxed in this analysis. Nevertheless, their sampled energetic roles differ (Figure 2). The free-cell centre is -14.838201 meV/HfO2 relative to its own PO+ well and lies below both adjacent samples. The clamped centre is +96.828183 meV/HfO2 relative to its different, boundary-relaxed PO+ well and lies above both adjacent samples. The low central region is therefore not present on this clamped band, even though the same group label remains. This does not establish that every possible clamped Pbcn branch has disappeared or that either centre is a stationary minimum or index-one saddle.

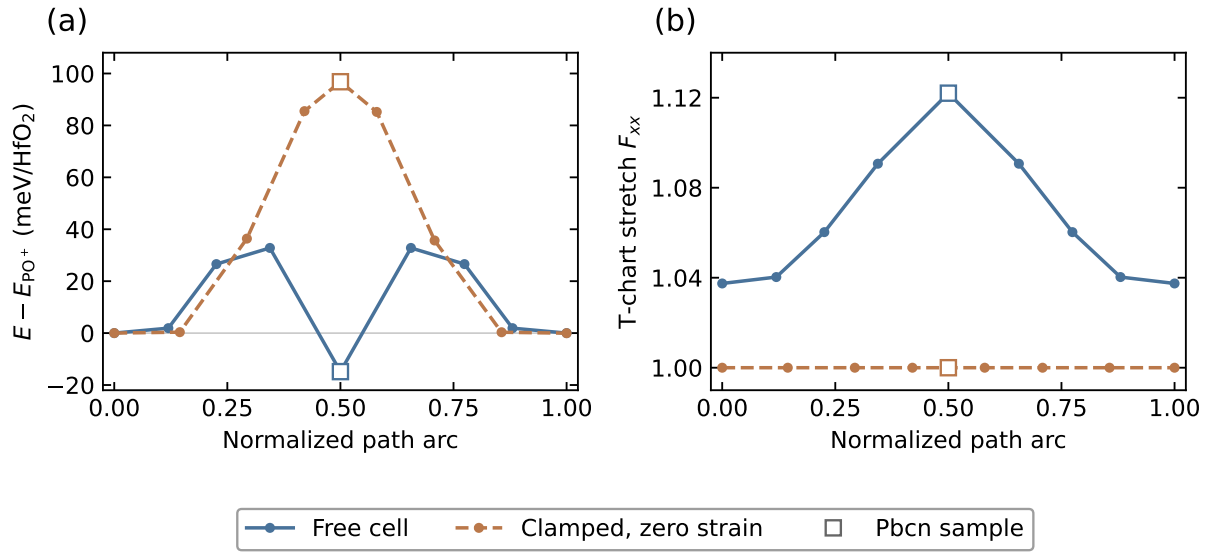


Figure 2. Preserving-pattern candidates with free cell (blue solid) and zero-strain clamp (orange dashed), using the unchanged electronic contract. (a) Native energies relative to each band's own PO+ initial well, not one common initial configuration. (b) Stretch F_{xx} in the common original T chart. Hollow squares identify the two Pbcn central samples; they are not certified TSs or stable intermediates. Nine existing samples per path are shown with straight connectors, with no smooth-MEP fit, statistical error band or independent replicate interpretation. The original free-cell nine-point sampled peak is 32.806023 meV/HfO2; the already measured sampling supplement raises its union maximum to 38.508448, which is not plotted as an additional optimized chain. Source CSV, the full tolerance sweeps and native-audit provenance are provided in the dated analysis case.

The geometries provide an associated boundary diagnostic. In the original T chart, the free-cell centre has $F_{xx} = 1.122022$; clamping fixes F_{xx} and F_{zz} to unity throughout the zero-strain band while allowing the original y direction and two corresponding tilts to respond. The atomic arrangement and allowed-cell response are both optimized along each band. Thus this comparison is neither a frozen-geometry energy difference nor an isolated strain derivative at one state. The six electronic input hashes are the same, but the mechanical boundaries, initial wells and original generalized metrics differ. Each band keeps its native normalized arc; equal abscissae are not a structural matching operation.

Clamping therefore changes more than the magnitude of a plotted energy: the same symmetry-labelled sample can lie in a low central region or at the sampled peak, depending on which deformation is permitted. The large free-cell x expansion shows why a phase label alone cannot specify its energetic role. The next comparison follows the actual atomic arrangements and the remaining open-cell response under one fixed boundary.

3.3. Distinct oxygen-shuffle routes couple to different lattice responses

The two switching paths reverse the polar displacement but do not perform the same oxygen rearrangement (Figure 3). To display that difference, we project the ordered, continuously lifted fractional displacements into one fixed original-T Cartesian metric. Three unit-normalized geometric oxygen-shuffle patterns, constructed from the tetragonal reference and its rotated variants, define signed amplitudes Q_x, Q_y, Q_z in Angstrom. A single initial periodic gauge is held over each whole chain; atoms are not rematched image by image. These amplitudes are geometric descriptors, not phonon occupations, assigned irreducible representations or the local joint coordinates Q_1, Q_2 used below. The companion polar descriptor is the mean oxygen displacement minus the mean hafnium displacement along the original polar z direction; it is not an electronic polarization.

Both zero-strain flips take this polar displacement from approximately +0.275 to -0.275 Angstrom. In the preserving route, however, the oxygen shuffle Q_x remains negative: it is -0.930 at the initial well, -0.799 at the sampled peak and -0.925 Angstrom at the final well. The reversing route takes the same initial negative shuffle through nearly zero at its sampled peak to +0.925 Angstrom. At +1% strain the preserving peak retains an even larger negative amplitude, -1.046 Angstrom, while the reversing peak again has nearly zero Q_x. The opposite barrier responses in Section 3.1 thus concern different atomic routes, even though both reverse the polar displacement and reach equivalent final-phase energies. Polar reversal alone does not label the path.

The T escape provides the complementary structural change. Instead of ending at the opposite polar well, it removes the initial negative Q_x and builds a positive Q_z, reaching 0.832 and 0.886 Angstrom at the two T endpoints. The polar displacement tends to zero. Along the sampled peak region the two patterns coexist: at zero strain the amplitudes are $Q_x = -0.723$ and $Q_z = 0.137$ Angstrom. The escape and switching candidates therefore share part of their initial distortion but differ in how it is removed or transferred into the final arrangement. A small displayed amplitude must not be mistaken for a high-symmetry crystal: at the reversing peaks the atomic displacement outside this triplet still has norm 1.392 and 1.895 Angstrom. The full-space and alternative-reference comparisons are retained in Appendices D.3 and F.1.

The allowed lattice response distinguishes these routes further. In the original T chart, x and z follow the substrate while y and its permitted tilts remain free. At the preserving sampled peak, the y stretch changes from -0.058% to +0.304% on increasing substrate strain. At the reversing peak it decreases from +1.741% to +1.417%; at the T escape peak it becomes more contractive, from -0.392% to -0.799%. These measurements connect the energy comparison to different coupled atomic-cell trajectories. They do not partition the energy into separate shuffle and elastic contributions: the coordinates relax together, and correlation of a stretch with a peak shift is not an isolated causal test.

The matched local relaxation calculations provide such a test at a more limited level (Figure 4). At each of five fixed pairs of local retained coordinates, we compare a frozen configuration, atomic-only release at its seed cell, and joint atomic/open-cell release. Across the five registered coordinates near the preserving peak, atomic relaxation alone lowers the energy by 11.016–12.087 meV/HfO2; the joint candidates are a further 0.201–0.284 meV/HfO2 lower. In the T neighborhood the corresponding ranges are 0.068–0.166 and 0.086–0.302 meV/HfO2. The large preserving lowering is therefore already present when the cell is held fixed at each seed, and cannot be attributed solely to release of the open lattice direction. Additional joint release includes atomic readjustment, so its energy difference is not a unique elastic contribution.

Reprojecting the same candidates identifies a structural distinction that the displayed oxygen amplitudes alone miss. Across the five preserving controls, the fixed-cell F-to-A displacement has a T-pattern component of 0.055–0.063 Angstrom and an orthogonal component of 0.303–0.316 Angstrom. The latter accounts for 95.9–97.1% of its squared norm. Oxygen contributes 80.8–81.2% of the total squared displacement, with a non-negligible Hf contribution; these are not energy percentages. The preserving Q_x moves toward zero while Q_z changes by less than 0.000335 Angstrom. By comparison, the T controls change Q_z by 0.020–0.036 Angstrom despite their much smaller atomic energy lowering. Thus the local energy contrast accompanies accommodation beyond the three plotted T patterns, rather than a common

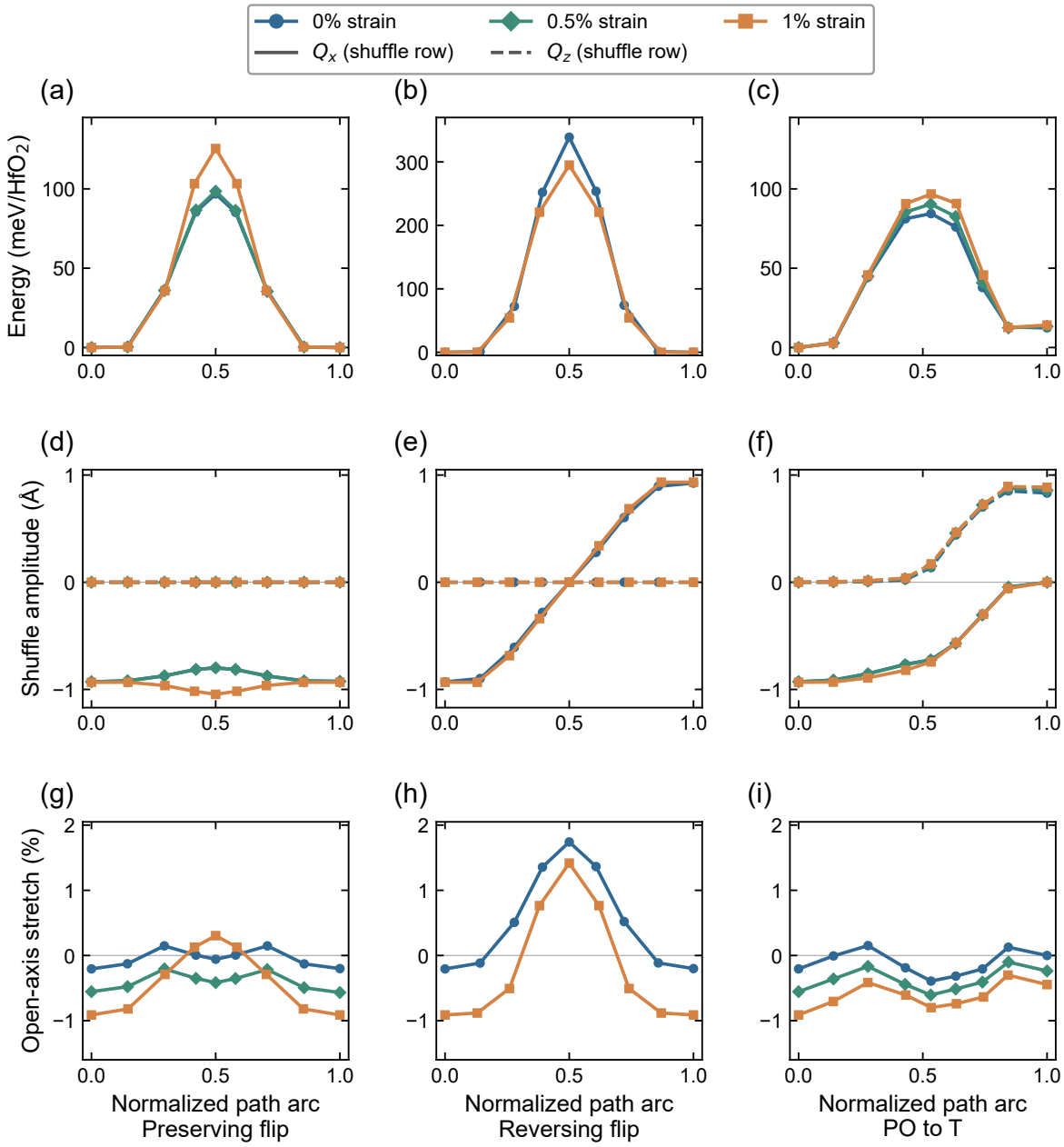


Figure 3. Energy, oxygen-shuffle coordinates and open-axis stretch on the preserving flip (left), reversing flip (middle) and PO-to-T escape (right). Blue circles and orange squares denote zero and +1% substrate strain; green diamonds add the independent +0.5% preserving and T paths. The reversing path was not evaluated at the midpoint. (a–c) Native energies relative to the matched initial well; the reversing column has a larger energy range. (d–f) Signed geometric Q_x (solid) and Q_z (dashed), in the same fixed reference metric and on common scales. (g–i) $100(F_{yy} - 1)$ in the original T chart, with x and z prescribed by the substrate. This stretch is not the Green-strain component or a strain energy. Each chain has its own normalized generalized arc; equal abscissae do not match structures between chains. Points are the nine actual images, including cached endpoints; straight connectors are not a fitted continuous MEP. All ten chains, including PO-to-M at the two training strains, and the omitted-coordinate norms are exported with the figure source.

change of one displayed amplitude. These observations concern two zero-strain neighborhoods; they do not by themselves determine the origin of the whole-path strain response.

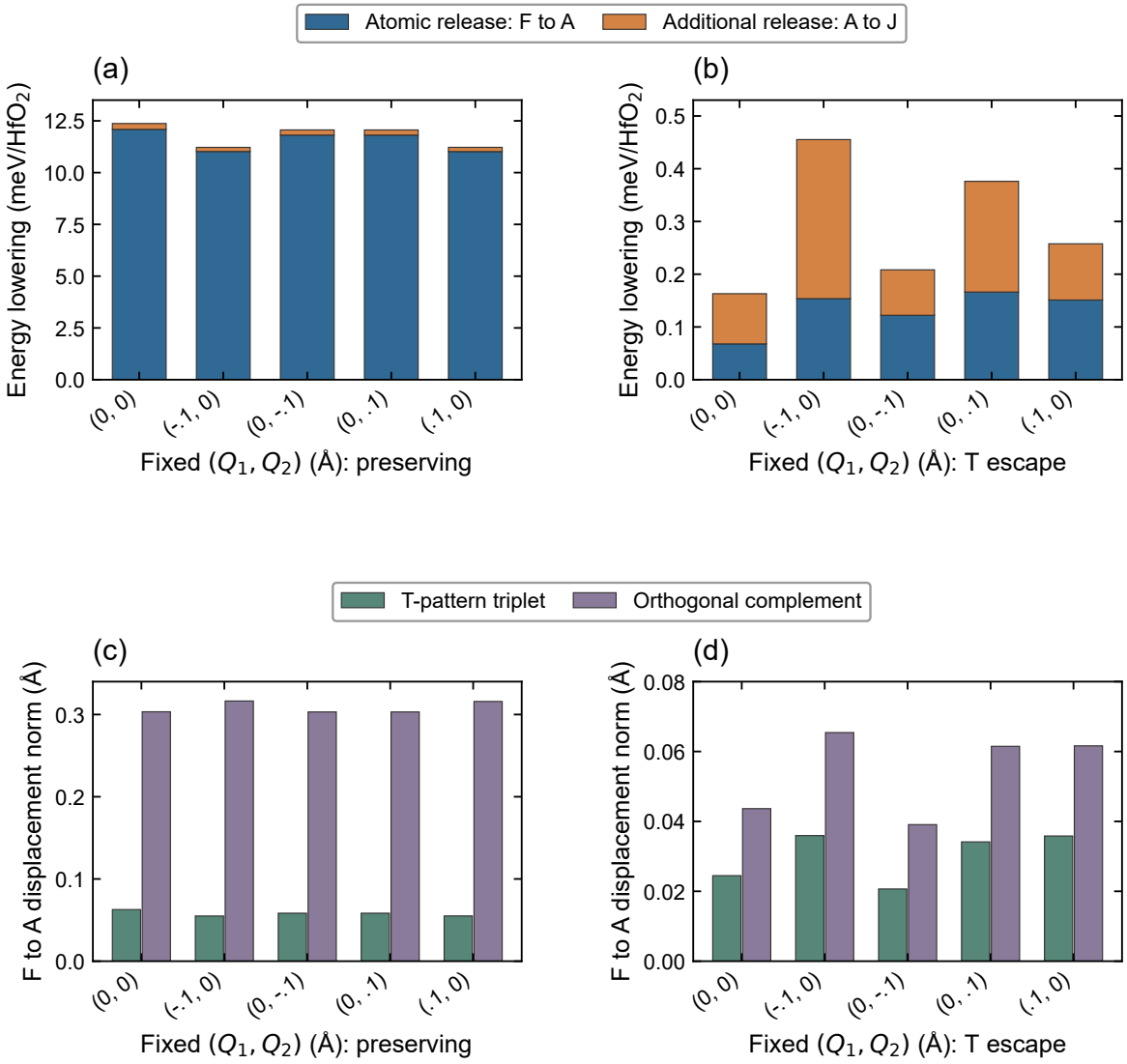


Figure 4. Local energy release and the atomic accommodation absent from the displayed T-pattern coordinates. (a,b) Measured energy lowering at the five prescribed retained coordinate pairs around the preserving and T sampled peaks, respectively, at zero substrate strain. Blue bars show $E_{\text{frozen}} - E_{\text{atomic}}$; orange increments show $E_{\text{atomic}} - E_{\text{joint}}$. (c,d) Norms of the same F-to-A translation-free displacement projected onto the three geometric T patterns and their complete orthogonal complement. These are lengths over the twelve-atom cell, not per-atom RMS displacements or energy fractions; the two norms do not add linearly. Within each row, the two panels deliberately use different vertical ranges. All ten physical points are included once; the two curvature probe steps are not duplicate energy observations. Q_1, Q_2 are the fixed joint-length coordinates of Section 2.3, not the geometric Q_x, Q_z of Figure 3. Each atomic-only treatment fixes its own frozen-Q seed cell. The total height is a local candidate-energy lowering, not a correction to the complete path barrier.

Together, these observations support a physical interpretation in terms of route-specific accommodation: the same substrate constraint acts on different oxygen rearrangements and on their different available lattice responses. The relaxation contrast further identifies internal atomic accommodation as an important contributor in the selected preserving neighborhood. It does not identify one mode as the sole cause of the

barrier change. In particular, the preserving origin still fails the original necessary sign condition after the two-coordinate release. Its shallow omitted oxygen-dominated direction and the bounded attempts to follow it are reported in Appendices E.7 and F.1. We therefore retain the measured relaxation effect without calling this neighborhood a stable conditional surface or retroactively lowering the path barrier.

The matched physical-pattern test across strain found only small residual accommodation at the three native PO-to-T sampled peaks (Figure 5). Fixed-cell atomic release lowered their energies by 0.058, 0.009 and 0.167 meV/HfO2 at 0, +0.5 and +1% strain, respectively. Additional joint release lowered them by 0.120, 0.077 and 0.073 meV/HfO2. The total local lowering was therefore 0.177, 0.086 and 0.240 meV/HfO2, with all six candidates meeting the same released-gradient target. No multi-meV residual lowering was observed under this four-observable restriction at these PO-to-T configurations. The native held values differ between strains, so these contrasts are not a fixed-coordinate strain derivative or a bound on a reoptimized path.

Small energy lowering did not require unchanged internal geometry. The F-to-J atomic displacement norms were 0.070, 0.022 and 0.097 Angstrom in the common original-T metric, compared with 0.036, 0.002 and 0.078 Angstrom for F-to-A. The additional A-to-J atomic readjustments were 0.041, 0.021 and 0.020 Angstrom, while the open-axis stretch increased by 0.141, 0.099 and 0.101 percentage points relative to F. Thus the joint contrast includes coupled atomic motion even when its energy is small. These same-definition T controls do not isolate the cost of adding one coordinate to Figure 4, whose retained subspaces are different. The following zero-strain preserving control addresses the same physical restriction, while its across-strain and stability coverage remains incomplete.

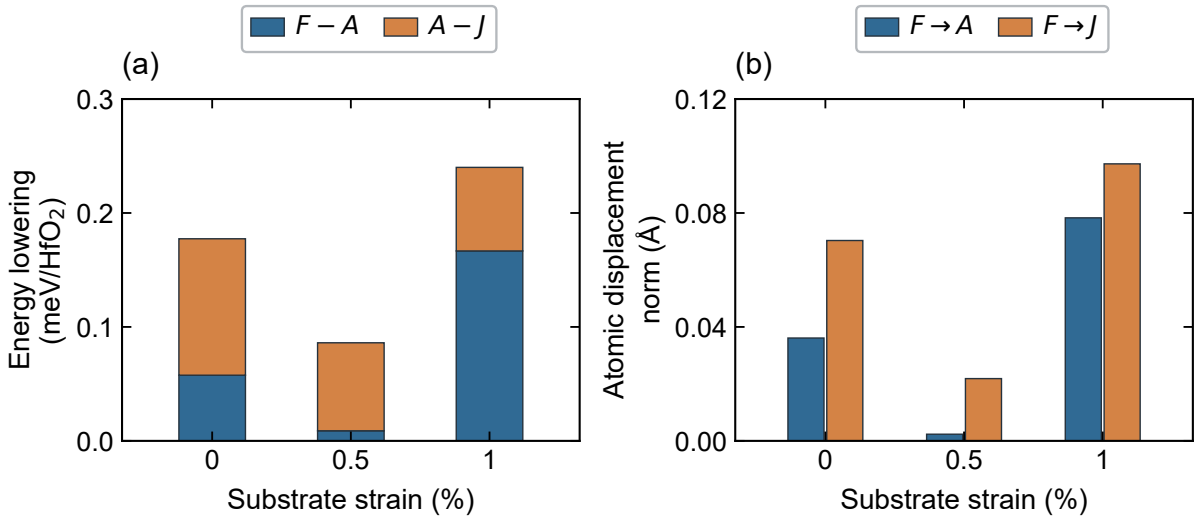


Figure 5. Residual accommodation at the three native PO-to-T sampled peaks with four physical observables retained. (a) Blue bars show the fixed-cell atomic lowering $E_F - E_A$; orange increments show the additional joint lowering $E_A - E_J$. The total height is $E_F - E_J$ in meV/HfO2. (b) Translation-free atomic displacement norms from F to A and F to J, using the original-T Cartesian metric over all twelve atoms. These are not per-atom RMS lengths, mass-weighted amplitudes or energy fractions. The definitions of Q_x, Q_y, Q_z and the structural polar z proxy are identical across the three mechanical conditions, but their native values are not equal. Within each triplet they are fixed, and A fixes its entire seed cell. Each bar represents one computed gradient candidate, not a statistical replicate; no uncertainty interval is inferred. Local lowering is not a corrected path barrier or a stability certificate. Source data and independent native replays accompany the calculation archive.

At zero strain, the same four-observable restriction left substantial preserving accommodation. Negative-start A lowered the native peak configuration by 13.599 meV/HfO2; J added another 0.018 meV/HfO2. The F-to-A atomic displacement norm was 0.477 Angstrom in the common original-T metric, with oxygen contributing 80.4% of its squared norm. Despite unchanged T-pattern amplitudes and z polar proxy, the transverse mean O-minus-Hf displacement increased by 0.122 Angstrom along y. This is a structural descriptor, not an electronic-polarization measurement or a minimum certificate. At +1%, both signed A starts exhausted twelve attempts near the fixed 0.5 Angstrom neighborhood boundary, with released gradients about 0.378 eV/Angstrom; their J energies remain unavailable.

The local gain survived a finite whole-path connection test (Figure 6). Reinserting J first moved the polygon's sampled maximum to a neighbor, at 85.500 meV/HfO2, with a residual of 0.1256 eV/Angstrom. Nine subsequent ordinary updates, with all seven interiors free, reached 0.094012 eV/Angstrom. The final sampled maximum returned to the central image at 83.166281 meV/HfO2, 13.661902 below the original band. The transverse y proxy remained 0.121074 Angstrom while Q_x changed only from -0.799 to -0.789 Angstrom at the peak. This is evidence that the mixed atomic accommodation can form a lower connected candidate, not merely an isolated low-energy point.

The comparison changes the candidate competition at zero strain. The original T escape peak is only 1.250343 meV/HfO2 above this new preserving candidate, rather than 12.411559 below the original one. The small new separation is not resolved as a physical ordering by the current sampling and force target. Neither a global MEP nor a stationary TS is certified, and the registered training labels are not replaced. The connection required 63 fresh interior SCFs and 55.857778 allocation core-hours; the earlier local-release and path costs remain additional.

3.4. The independent strain test resolves different sampled responses

Both preregistered +0.5% bands met the same ordinary NEB criterion: the preserving flip reached 0.079773 eV/Angstrom and PO-to-T reached 0.080825 eV/Angstrom. Their nine-image sampled peaks were 98.530877 and 90.418178 meV/HfO2, respectively (Figure 7). Relative to zero strain, the preserving peak increased by 1.702694, whereas the T escape peak increased by 6.001554 meV/HfO2. The common starting well, electronic inputs, ordered path definitions and stopping criterion were kept as registered.

The selected-pair contrast, T minus preserving, was -8.112699 meV/HfO2 at +0.5%, compared with -12.411559 at zero and -28.660781 at +1%. Thus the midpoint contrast improved by 4.298861 meV/HfO2 relative to zero, before worsening at the larger imposed strain. It remained negative: escape still had the lower sampled peak in this tested pair. This changes the interpretation from a uniformly unfavorable strain trend to a nonmonotonic sampled competition. Only the two preselected channels were computed at +0.5%; their comparison does not establish the complete four-channel ordering at that condition.

The frozen B0 interpolation was accurate for T escape at this resolution but overpredicted the preserving peak by 12.595430 meV/HfO2 (Table 3). Its predicted pair contrast, -20.536170, missed the measured value by -12.423471 meV/HfO2. Conversely, both endpoint-only B1 forecasts were closer for preserving than B0 but less accurate for T escape. B1-follow had the smallest maximum error across the two paths, 5.004639 meV/HfO2. A single linear response inferred from the two outer strains therefore did not describe both registered channels equally well.

Table 3. Frozen physical-strain forecast errors, prediction minus DFT, in meV/HfO2. Pair error is the error in T minus preserving. All three models cover both targets; there is no post-label calibration.

Model	Preserving error	T error	Max. absolute error	Mean absolute error	Pair error
B0 midpoint	12.595430	0.171959	12.595430	6.383694	-12.423471
B1 fixed absolute peak	-2.933378	-7.232239	7.232239	5.082809	-4.298861
B1 final-well following	-1.667843	-5.004639	5.004639	3.336241	-3.336796

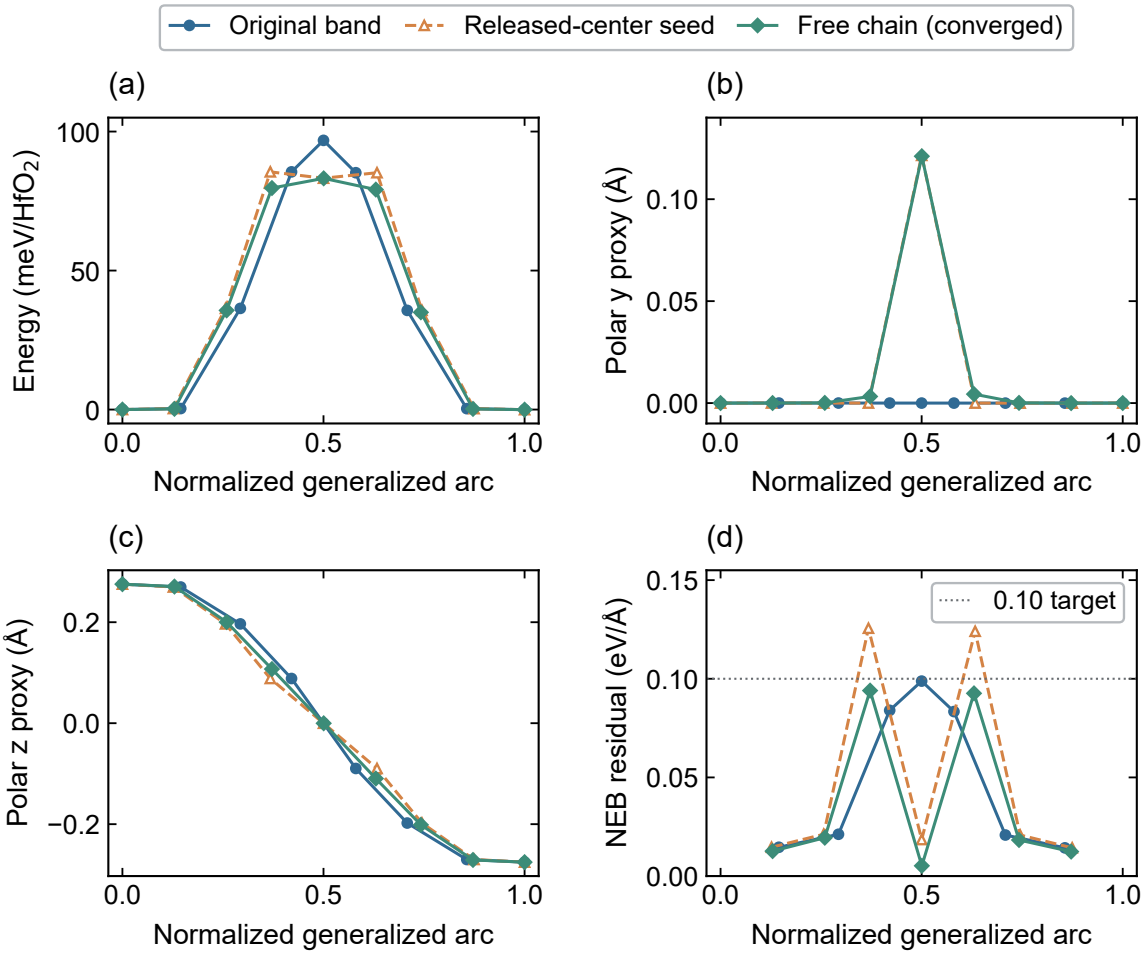


Figure 6. Connecting local atomic accommodation to the complete zero-strain preserving band. Blue circles denote the original band, hollow orange triangles its released-center seed and green diamonds the final ordinary band. (a) Native sampled energies relative to the unchanged initial well. (b,c) Mean O-minus-Hf displacements along original y and z, respectively; these are structural proxies, not electronic polarization. (d) Seven interior NEB residuals with the 0.10 eV/Angstrom target. Endpoint residuals are undefined and omitted, not set to zero. Each polygon has its own normalized generalized arc; straight connectors do not resolve a smooth MEP. The final chain retains the original endpoint caches and has no local-pattern constraints. One prescribed seed was tested, not statistical replicates or an acceleration comparison. All 63 fresh SCFs accompany the native archive and were independently reparsed.

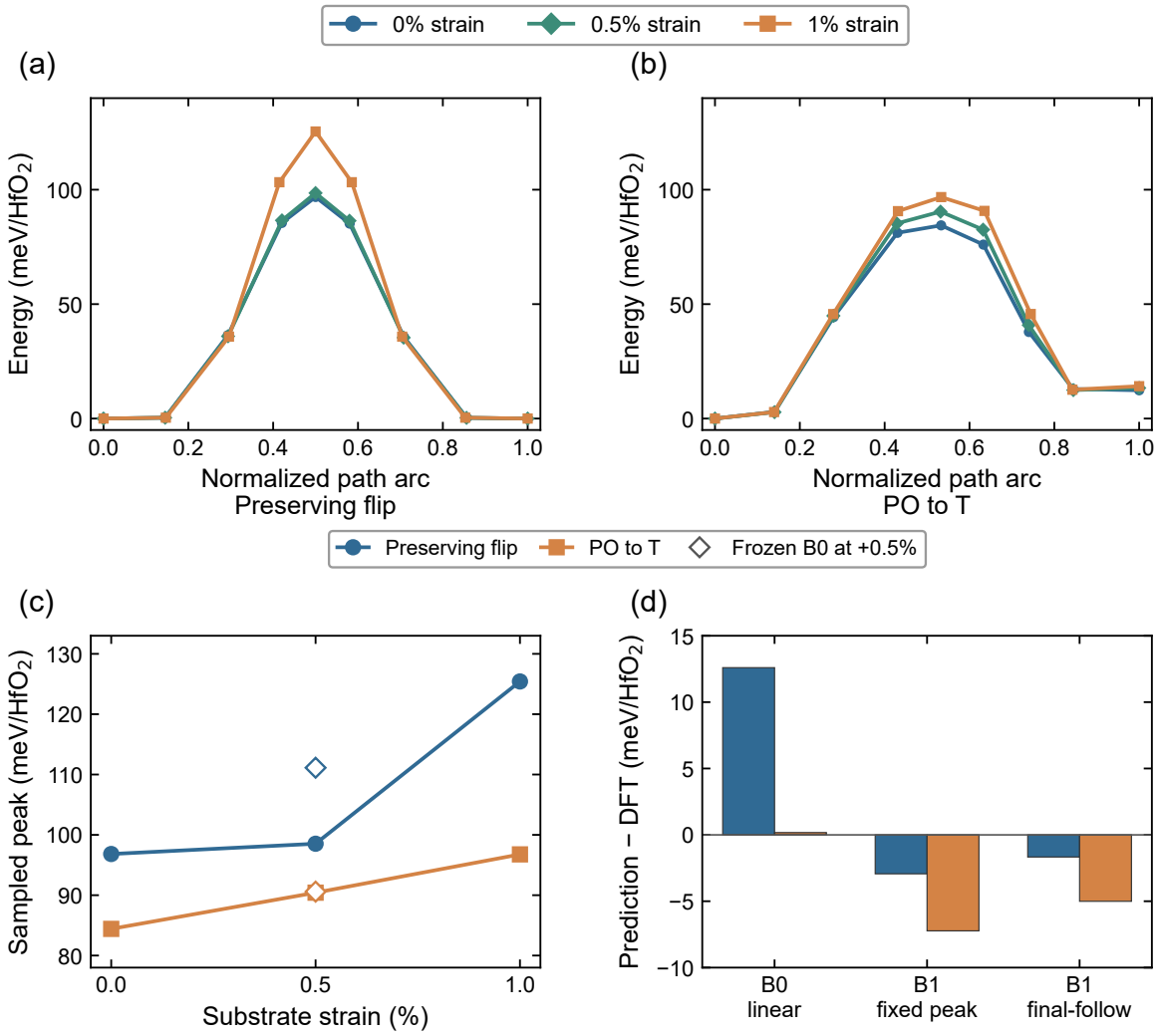


Figure 7. The independently evaluated +0.5% mechanical condition. (a,b) Actual nine-image preserving and PO-to-T profiles alongside their zero/+1% training bands; blue, green and orange encode strain. (c) Sampled peaks versus strain, with blue circles for preserving and orange squares for T escape. Hollow diamonds are the B0 midpoint forecasts frozen before target-interior evaluation. (d) Signed errors of all three numerical forecast arms. Channel colors in (c,d) differ from the strain colors in (a,b), as specified by the separate legends. Straight connectors are visual guides through measured samples; no smooth-MEP or strain fit is introduced. Both bands use the same target PO+ cache and the same electronic parameters. No uncertainty interval is inferred from the force criterion.

The structural check preserves the distinction between the two routes (Figure 3). At the +0.5% preserving peak, $Q_x = -0.798917$ Angstrom and Q_z remains nearly zero, close to the zero-strain pattern and different from the larger negative Q_x at +1%. The T peak instead retains mixed $Q_x = -0.725241$ and $Q_z = 0.150244$ Angstrom, intermediate in Q_z between the two training conditions. Their open-y stretches are -0.416145% and -0.605986%, respectively. The midpoint data therefore retain the route identities while exposing different sampled energy and lattice responses; no new phase or bifurcation is assigned from these descriptors.

This test used the registered zero-strain polygons as warm starts after affinely imposing the target plane and distributing the relaxed endpoint corrections. The preserving seed already passed the ordinary criterion after its first seven interior SCFs, with no optimizer update; T required two updates and 21 interior SCFs. These are completed fixed-protocol tests, rather than independent searches over starting paths. Their finite image resolution, stopping tolerance and history do not separate genuine continuous-path curvature from numerical path-relaxation effects. Accordingly, the measured deviation from B0 is reported for the declared sampled observable, without claiming a transition-state error bound.

A separate continuation from the +1% training band tested initialization sensitivity at the same +0.5% boundary, with identical cached endpoints and electronic settings. For PO-to-T, this control also required two ordinary NEB updates and 21 interior SCFs. It reached a residual of 0.082374 eV/Angstrom and a sampled peak of 90.286243 meV/HfO2, only 0.131935 meV/HfO2 below the zero-strain-start result. The peak geometries were not identical: their Q_x values were -0.741484 and -0.725241 Angstrom, and open-y stretches were -0.502644% and -0.605986%, respectively. Thus the T sampled energy was insensitive to this particular change of starting band, despite a remaining geometric difference. This finite control is not an error bound on the continuous barrier or evidence of a unique MEP.

The preserving control was substantially more sensitive (Figure 8). It required 26 updates and 189 fresh interior SCFs to reach 0.099893 eV/Angstrom, with a sampled peak of 109.351721 meV/HfO2. This is 10.820844 above the registered zero-strain-start midpoint, comparable with its 12.595430 meV/HfO2 deviation from the frozen B0 forecast. The peak's Q_x changed from -0.798917 to -0.881752 Angstrom and its open-y stretch from -0.416145% to -0.046946%. The paired T-minus-preserving contrast consequently changed from -8.112699 to -19.065478 meV/HfO2. The previously observed 4.298861 meV/HfO2 midpoint improvement therefore does not survive this initialization control as a path-independent response. The two sampled outcomes do not distinguish a distinct continuous branch from residual relaxation and finite-discretization effects. They also do not assign the energy difference uniquely to the atomic pattern or cell. Both original prospective labels and all frozen forecast scores remain unchanged. These two initialization controls add 251.048889 allocation core-hours, including the T control already described, not a new cost for re-exporting it.

All 138 frozen forecast records remain: six numerical predictions and 132 unavailable physical-transfer records across the two cases. The local reduced models had not supplied a valid transfer to this physical control, so this test establishes no predictive advantage for a branch-adapted mode model. The bounded physical test is complete; its outcome does not initiate model reselection. The 28 path SCFs and eleven endpoint SCFs consumed 47.289888 native-call proxy core-hours and 48.808889 allocation core-hours together. Earlier training, curvature and local-model preparation remain additional.

An independent prediction advantage has not yet been established. The prospective T local test supplied eight independent candidate energies, with coordinates and model predictions frozen beforehand. All eight reached the prescribed released-gradient target in eighteen native SCFs. The tensor-quadratic interpolant had a maximum origin-aligned error of 0.089077 meV/HfO2, versus 0.129769 and 0.131911 for the two original curvature models. The original models had smaller unaligned maxima, however; the ranking depended on the declared energy-zero convention. This supports local candidate-energy description on that tested domain, not a unique benefit of an adaptive reference.

The separately revised preserving test also completed all eight candidates with both gradient tests in 24 native SCFs. On the seven targets shared by the four new family/step arms, adding the retained-direction nonlinear terms slightly reduced RMSE but increased maximum error at both steps. For example, at the

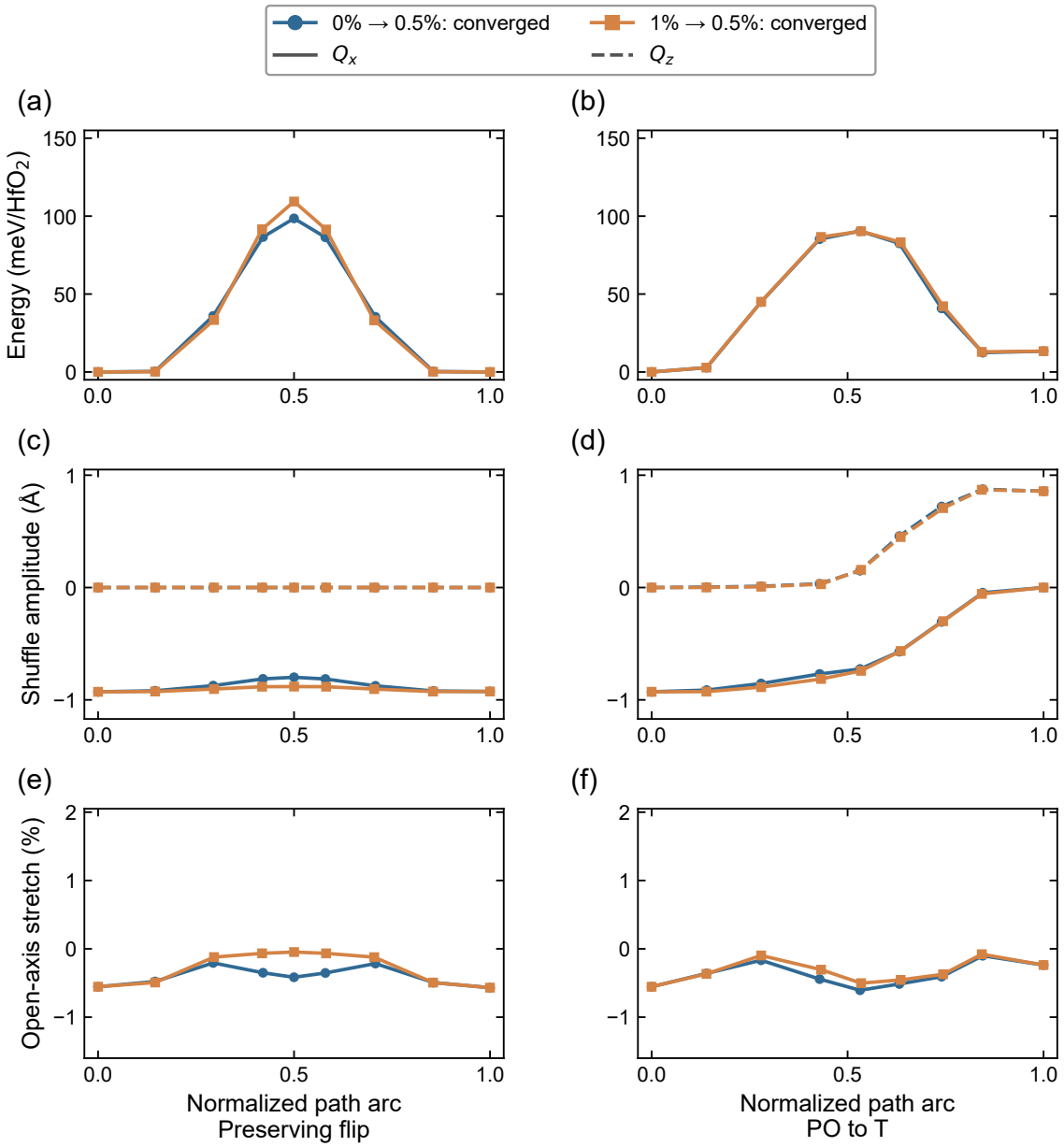


Figure 8. Two ordinary-converged initializations at the same +0.5% mechanical boundary. Blue circles start from zero-strain bands and orange squares from +1% bands; both use the identical target endpoint caches. Left and right columns show preserving reversal and T escape, respectively. (a,b) Native sampled energy profiles. (c,d) Signed Q_x (solid) and Q_z (dashed) in the same original-T metric. (e,f) Allowed y stretch in the original chart. There are nine actual images per band, each with its own normalized arc. The T energy agreement does not imply identical structures; the larger preserving difference prevents attributing the registered midpoint nonlinearity uniquely to intrinsic strain response. This is a follow-up sensitivity test, not a second independent prediction cohort, an exhaustive branch search or a continuous-barrier error bound.

first step the harmonic/nonlinear maxima were 0.033195/0.035625 meV/HfO2 and their RMSEs were 0.027153/0.024916. These small local errors do not establish a physically resolved model advantage or remove the original two-coordinate refusal. Coverage, all older controls, unsuccessful training starts and actual preparation costs remain in Appendices E.8–E.9 and F.2. No independent labels were used to refit these predictions.

4. Discussion: mechanical control acts on the route as well as the wells

The central finding is a mismatch between lowering one switching path and improving the competition available to the polar crystal. Under the examined +1% substrate intervention, the high reversing candidate becomes lower while the already lower preserving candidate becomes higher. The easiest sampled escape candidate rises less than the preserving candidate, so the escape-minus-switching contrast becomes more negative. Within this declared candidate set, +1% strain therefore does not produce the favorable switching-versus-escape response sought here. Selecting only the decreasing reversing barrier would give the wrong account of this competition.

The independent midpoint refines this conclusion. At +0.5%, the selected escape-minus-switching contrast improves relative to zero strain but stays negative; at +1% it is more negative. The registered sampled competition is therefore nonmonotonic over these three conditions. The preserving peak's midpoint deviation from B0 contrasts with the nearly interpolating T peak, while the signed mode patterns retain their route identities. This favors examining route-dependent atomic and lattice accommodation over an interpretation based only on endpoint well shifts. The warm-started ordinary bands do not identify a physical bifurcation or isolate the origin of this finite-resolution nonlinearity. The completed reverse-continuation control supports the T sampled peak against one change of initialization, but changes the preserving peak by 10.820844 meV/HfO2. This exceeds the registered 4.298861 meV/HfO2 midpoint improvement of the paired contrast. Consequently the nonmonotonic registered contrast is not an initialization-independent strain-response law. In particular, nearly equal T peak energies at different oxygen amplitudes and open-axis stretches do not isolate a unique atomic or elastic cause.

The mode and strain results explain why a single polar coordinate or an endpoint phase label is insufficient. The two flips reverse essentially the same polar displacement but preserve or reverse a signed oxygen shuffle, respectively. Escape to T instead exchanges the initial pattern for its differently oriented tetragonal counterpart. These distinct rearrangements have different open-axis responses under the same imposed substrate change. The equal final-well responses of the two flips cannot account for their opposite peak shifts. Their path interiors must be resolved to interpret the mechanical response.

Mechanical boundary conditions also determine which accommodation is available. The free-cell preserving path develops a large x expansion near its low central region; that expansion is excluded by the clamped ensemble, where the same Pbcn-labelled central sample has a high energy relative to its own initial well. This is a boundary-dependent role of a configuration family, not a universal energetic meaning of a space-group label. A comparison of relaxed endpoints must therefore retain the mechanical ensemble as part of the definition of the pathway.

The local treatment comparison adds evidence beyond geometric correlation. The large energy lowering near the preserving peak already occurs during atomic release at each fixed seed cell, whereas the T neighborhood has a much smaller atomic lowering. Thus lattice freedom is not the sole source of relaxation in the preserving neighborhood. This result supports a coupled atomic-cell interpretation without assigning a unique energy fraction to one mode. The paired structural data place most of this fixed-cell accommodation outside the three displayed T-pattern directions, with both oxygen and Hf motion. This directly limits the mechanistic information carried by those geometric projections. It also explains why a visually compact two-coordinate plot can omit energetically important accommodation. The two-coordinate release itself remains local; the separate four-observable intervention and subsequent free-interior connection provide the whole-path test.

The across-strain T control further separates a compact geometric description from an assertion that all omitted coordinates are already equilibrated. Retaining the displayed physical patterns and polar proxy left measurable atomic and open-cell readjustment, but the measured local lowering stayed below 0.25 meV/HfO2 at all three PO-to-T peaks. This provides no evidence for a large unaccounted complementary release at those configurations under the tested restriction. It does not certify their full stability, constrain every alternative T route, or explain the preserving response by subtraction. At zero strain, the same-definition preserving intervention instead exposes a mixed atomic reorganization with a transverse structural polar component that is invisible to the held z proxy and T-pattern amplitudes. Relaxing the reconnected band retained that transverse component and lowered its sampled peak by 13.661902 meV/HfO2. A local force pass on the original high-symmetry candidate had therefore not exhausted the available atomic accommodation. The resulting near-degeneracy with T escape also removes a robust claim of escape preference at zero strain. This does not prove that one transverse coordinate alone causes the reduction: several atomic directions were released together. Its across-strain behavior remains incomplete because the bounded +1% preserving releases did not supply converged A/J candidates; the small T releases cannot fill that gap.

The model tests place an important limit on the methodological claim. A complete change of coordinates leaves the same quadratic response unchanged, and the independent candidate-energy tests do not show a metric-independent accuracy gain for the added nonlinear family. These outcomes do not negate the measured channel competition or structural contrast. They instead separate the material result from a stronger, currently unsupported claim that one particular reduced model predicts it better. The contribution is the connected comparison of competing routes, their mechanical responses, a controlled local relaxation test and a prospectively frozen physical-strain test. A new predictive-method claim would require additional independent evidence beyond the completed comparison.

The quantitative scope is periodic twelve-atom hafnia at zero field and temperature under the fixed ABACUS/PBE contract. The four ordered candidates are not an exhaustive search, ordinary-converged sampled peaks are not certified stationary transition states, and the strained T/M separation of 0.464068 meV/HfO2 is not a resolved physical ordering. The study does not infer device switching rates, retention times, domain-wall nucleation or finite-temperature free energies. The registered +0.5% test is now complete, with both successful path outcomes and all model abstentions retained. The resulting evidence supports a material pathway comparison and explicit limits on response transfer, rather than a validated new predictive reduction method.

5. Conclusions

Mechanical control of hafnia must be assessed through the competition between complete structural routes from the same polar well. Within the original registered clamped candidate set, +1% strain lowers the high reversing switching peak while raising the already lower preserving peak more than the lowest escape peak. Selecting the decreasing switching profile alone would therefore misrepresent the available competition. The independent +0.5% test adds a different response: the selected escape-minus-switching contrast improves relative to zero strain but stays negative. However, the later initialization test shifts the preserving midpoint by 10.820844 meV/HfO2 while leaving the T peak nearly unchanged. The registered nonmonotonic trend cannot therefore be interpreted as an intrinsic, initialization-independent strain law.

The structural comparison identifies the atomic and lattice accommodation that distinguishes the routes. Reversal of the polar displacement can accompany preservation or reversal of a signed oxygen shuffle; escape to T transfers the initial distortion into a differently oriented pattern. Each route uses a different allowed lattice response. Equal final-well shifts cannot explain the opposite switching-peak changes, so the path interior must enter the interpretation. Clamping also changes the sampled energetic role of a Pbcn-labelled central configuration while excluding the large expansion seen on the free-cell path. Matched local controls show that internal atomic relaxation already recovers most of the preserving

neighborhood's energy lowering at fixed seed cells. These observations support coupled atomic and lattice accommodation as the physical basis for interpreting the route differences, with no unique mode-by-mode energy partition. The four-observable preserving intervention and free-chain connection further recover a 13.661902 meV/HfO2 lower zero-strain sampled peak while retaining transverse atomic accommodation. This brings reversal and examined T escape close enough that their physical ordering remains unresolved. The original displayed patterns and force criterion alone were insufficient to make that competition robust.

The independent comparison further shows that good interpolation for T escape does not ensure a comparable description of preserving reversal. The tested local reduced models establish no additional physical-response prediction advantage. The resulting contribution is a reproducible connection between channel competition, ordered structural descriptors, permitted strain and controlled relaxation, together with a measured test of response transfer. Its quantitative scope is the declared zero-field, zero-temperature sampled paths under one ABACUS/PBE contract; it does not determine device rates or an exhaustive transition network.

Data and code availability

VARNEB source code is available from the project repository. The calculation geometries, native-output exports, figure source tables, projection and relaxation analyses, frozen forecasts and replay scripts supporting this article are included in the same repository and linked throughout the appendices. The calculation and data version used for the completed independent test is Git revision dc328d50. The later two-channel initialization control and its complete native outputs are archived separately in the E269 case and the lower connected zero-strain candidate in the E273 case, whose terminal data first entered Git revision 902da0d9. Neither follow-up replaces the original forecast labels. The reading manuscript and its TeX wrapper are in paper/VARNEB_JCTC; the scalar tables and main figures can be regenerated without launching a DFT calculation. Executables and pseudopotential/orbital binaries are not redistributed with these exports; their exact identities are recorded by the archived input hashes.

References

1. Liu, S.; Hanrahan, B. M. Effects of growth orientations and epitaxial strains on phase stability of HfO2 thin films. Physical Review Materials 2019, 3, 054404. DOI: 10.1103/PhysRevMaterials.3.054404.

2. Behara, S. S.; Van der Ven, A. Ferroelectric HfO2 and the importance of strain. Physical Review Materials 2022, 6, 054403. DOI: 10.1103/PhysRevMaterials.6.054403.

3. Delodovici, F.; Barone, P.; Picozzi, S. Trilinear-coupling-driven ferroelectricity in HfO2. Physical Review Materials 2021, 5, 064405. DOI: 10.1103/PhysRevMaterials.5.064405.

4. Qi, Y.; Singh, S.; Rabe, K. M. Polarization switching in ferroelectric HfO2 from first-principles lattice mode analysis. Physical Review B 2025, 111, 134106. DOI: 10.1103/PhysRevB.111.134106.

5. Qi, Y.; Rabe, K. M. Competing Phases of HfO2 from Unstable Flat Phonon Bands of an Unconventional High-Symmetry Structure. Physical Review Letters 2025, 135, 046101. DOI: 10.1103/9759-kp38.

6. Lee, I.; Lee, W.; Yu, J. Strain-tuned ferroelectric transitions in HfO2: role of X_2^- mode in ferroelectric instabilities. npj Quantum Materials 2026, 11, 34. DOI: 10.1038/s41535-025-00841-9.

7. Zhou, S.; Zhang, J.; Rappe, A. M. Strain-induced antipolar phase in hafnia stabilizes robust thin-film ferroelectricity. Science Advances 2022, 8, eadd5953. DOI: 10.1126/sciadv.add5953.

8. Fan, D.; Zhu, T.; Liu, S. Unexpected density functional dependence of the antipolar Pbcn phase in HfO2. npj Computational Materials 2025, 11, 140. DOI: 10.1038/s41524-025-01647-w.

9. Ma, L.-Y.; Liu, S. Structural Polymorphism Kinetics Promoted by Charged Oxygen Vacancies in HfO2. Physical Review Letters 2023, 130, 096801. DOI: 10.1103/PhysRevLett.130.096801. The mapping comparison uses the public arXiv:2204.09374v3 supplement.

10. Lee, H.-J.; Kumar, P.; Go, K.-J.; et al. Bulklike Costless Domain Walls Driven by Phonon Pair Condensation in HfO2. Physical Review X 2026, 16, 031041. DOI: 10.1103/63lf-k7zs.

11. Henkelman, G.; Jónsson, H. Improved tangent estimate in the nudged elastic band method for finding minimum energy paths and saddle points. J. Chem. Phys. 2000, 113, 9978–9985. DOI: 10.1063/1.1323224.

12. Sheppard, D.; Xiao, P.; Chemelewski, W.; Johnson, D. D.; Henkelman, G. A generalized solid-state nudged elastic band method. J. Chem. Phys. 2012, 136, 074103. DOI: 10.1063/1.3684549.

13. Qian, G.-R.; Dong, X.; Zhou, X.-F.; Tian, Y.; Oganov, A. R.; Wang, H.-T. Variable cell nudged elastic band method for studying solid–solid structural phase transitions. Computer Physics Communications 2013, 184, 2111–2118. DOI: 10.1016/j.cpc.2013.04.004.

14. Larsen, A. H.; Mortensen, J. J.; Blomqvist, J.; et al. The atomic simulation environment–a Python library for working with atoms. Journal of Physics: Condensed Matter 2017, 29, 273002. DOI: 10.1088/1361-648X/aa680e.

15. Li, P.; Liu, X.; Chen, M.; Lin, P.; Ren, X.; Lin, L.; Yang, C.; He, L. Large-scale ab initio simulations based on systematically improvable atomic basis. Computational Materials Science 2016, 112, 503–517. DOI: 10.1016/j.commatsci.2015.07.004.

16. Chen, M.; Guo, G.-C.; He, L. Systematically improvable optimized atomic basis sets for ab initio calculations. Journal of Physics: Condensed Matter 2010, 22, 445501. DOI: 10.1088/0953-8984/22/44/445501.

17. Perdew, J. P.; Burke, K.; Ernzerhof, M. Generalized Gradient Approximation Made Simple. Physical Review Letters 1996, 77, 3865–3868. DOI: 10.1103/PhysRevLett.77.3865.

Appendix A. Reproducibility and scope

The archived cases provide frozen numeric observations, input/log/geometry checksums, continuous ordered lifts, analysis commands and figure CSVs. Raw production logs remain read-only on HF. Earlier incomplete snapshots remain dated evidence rather than being overwritten by final values. No external paper figure or unpublished author dataset is copied into this draft.

The eight clamped training bands, matched local controls, local independent candidate-energy tests and two-path physical-strain test are complete. Their detailed outcomes, including failed branch trials and unavailable forecasts, remain in the appendices and archived cases. The interpretation in the main text concerns those measured sampled paths. Stable connected conditional surfaces and stationary transition-state indices were not established and are not assumed in its material conclusions.

The main-figure and scalar-table replay commands read the archived data without invoking a calculator. Full byte-bound physical inputs and native outputs remain associated with the calculation records; the three visible endpoint features and complete forecast panel preceded both target-interior calculations. No physical error interval is inferred from an ordinary NEB residual. The appendices provide the extended evidence within this article, without a separate supplementary manuscript.

Appendix B. Why a constraint need not increase a relaxed-endpoint barrier

Consider the dimensionless illustrative potential

$$\mathcal{V}(x, y) = (x^2 - 1)^2 + (1 + x^2)y^2.$$

The coordinates and energy scale are arbitrary; this is not a fitted hafnia potential, phonon-coordinate model, new DFT point or independent prediction. The unrestricted minima are $(x, y) = (\pm 1, 0)$ with energy zero. Every continuous path connecting them crosses $x = 0$, where $\mathcal{V} = 1 + y^2 \geq 1$, and the path $y = 0$ attains this bound. The ideal unrestricted barrier is therefore one.

With the prescribed constraint $y = 1$, the relaxed endpoints instead become $x = \pm 1/\sqrt{2}$, with energy $7/4$. Their constrained bottleneck at $x = 0$ has energy two and negative curvature in the allowed x direction. The activation barrier is consequently $2 - 7/4 = 1/4$, despite the higher absolute bottleneck energy. The nonzero y gradient is a prescribed-coordinate reaction, not failure of allowed-coordinate stationarity. The unrestricted endpoints are absent from the constrained path space, so the common-endpoint inequality has not been violated. This elementary distinction motivates reporting both well and bottleneck responses; it supplies no favorable hafnia window or methodological novelty. The analytic derivatives, stationary-point curvatures and a continuous family of fixed-y examples are checked in the analytic checks.

Appendix C. Free-cell candidates and finite sampling checks

These dated free-cell observations establish the source correspondences and diagnostic limits, not the current state of the clamped training matrix. The qualified G1 review subsequently permitted G2; Table 1 in the main text contains the completed matched-boundary comparison. Free-cell energies are not reused as clamped energies. The following source records and four figures are retained rather than overwritten by later calculations.

C.1. The formation band and the shared polar starting well

The repaired T-PO band passes the ordinary residual criterion with 0.0598816 eV/Angstrom. Its ten frozen images reproduce complete same-contract energies, forces and stresses; its endpoints are unchanged. The discrete T-to-PO maximum is 33.889 meV/f.u. Reversing the thermodynamic view, without reoptimizing or remapping the band, gives 115.210 meV/f.u. for PO-to-T. Their difference equals the endpoint energy change, -81.321 meV/f.u. This is the first residual-passed candidate edge, not a sampling-resolved absolute activation barrier or a full-variable TS. (Numerical source)

All four candidate observations share the ordered PO+ energy -9783.249675811956 eV/cell. The following table uses the complete frozen terminal update audited by 18:12 on 9 October, not rounded live-log values:

Candidate from PO+	Source job/step	Residual	Maximum	Pass
PO-to-T, reverse view	28300425/6	0.059882	115.210	Yes
PO-to-M, refined	28392675/1	0.096964	82.680	Yes
T-pattern-preserving flip	28319570/69	0.099372	32.806	Yes
T-pattern-reversing flip	28319571/45	0.094023	392.823	Yes

Table A1. Four ordinary-residual-passed candidates, not an error-bounded activation-barrier ranking. Snapshot steps and times differ; M has twelve images after the separate sampling refinement. Residuals are in eV/Angstrom and sampled maxima in meV/f.u.; maxima include the endpoints and use four formula units per cell. The two flip labels denote registered geometric operations, not certified distinct MEPs

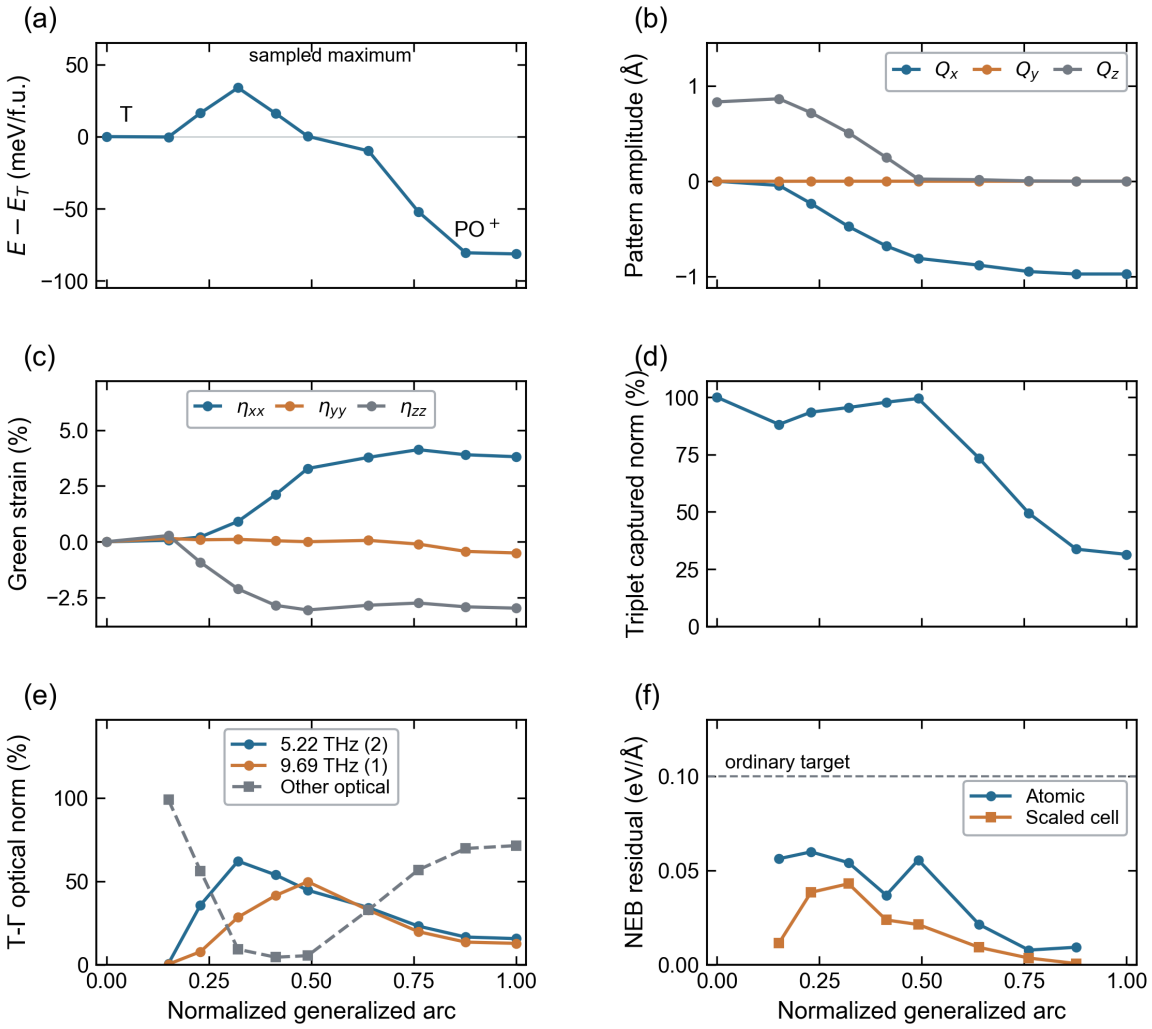


Figure A1. Audited discrete energy, geometric pattern amplitudes, Green strain, reference-subspace weights and ordinary residuals along T-to-PO. Connections join samples, not an independently validated smooth MEP. Reference phonons are not local bottleneck eigenvectors or mode energy contributions; fixed endpoints have no NEB residual. Complete captions, CSV and reproduction instructions are in the figure bundle.

or assignments to published irreps. The forty-record terminal audit includes cached endpoints and reused records, not forty new or independent SCFs. The earlier nine-image M and incomplete reversing observations remain dated evidence in their original archives.

PO-to-M terminated normally at the unchanged force threshold. Its ordered M endpoint is P2_1/c at all three declared symmetry tolerances. The refined sampled forward/reverse barriers are 82.680/154.065 meV/f.u., with unchanged reaction energy -71.385 meV/f.u.; their difference follows from the common endpoint energies. Its initial nine-image maximum was 71.582 meV/f.u.; the source's physical tangential force at that sampled peak was -0.383495 eV/Angstrom. This earlier diagnostic prompted the finite sampling check and subsequent ordinary refinement described below. Neither the old nor the refined residual pass alone establishes a stationary bottleneck.

C.2. Nonpolar snapshots do not uniquely identify a switching mechanism

The earlier preserving snapshot at step 14 had a Pbcn central sampled maximum across the declared 0.001/0.01/0.05 Angstrom symmetry tolerances. The later profile split into two sampled local peaks. At the ordinary-converged step 69, images 3 and 5 are approximately 32.806 meV/f.u. above PO+. Both are Pca2_1 at those tolerances. The central image remains Pbcn but is now below PO+, -14.838 meV/f.u., with an atomic/cell NEB residual of 0.062019/0.099372 eV/Angstrom. Transverse stability is unmeasured. It is neither the current highest image nor a certified stable intermediate. A phase label or an earlier central peak cannot therefore fix the eventual bottleneck location. If that lower nonpolar branch is stable, it also supplies a potential escape route from the polar well. M/T comparisons alone could then overstate protection. This coverage limitation is retained explicitly; the registered branch/stability tests must resolve its interpretation before a global escape-resistance claim, without adding a post hoc endpoint matrix.

At the two side peaks, the physical generalized tangential forces are approximately +0.242486 and -0.242483 eV/Angstrom, despite small atomic/cell NEB residuals at those images. Those forces distinguish the ordinary perpendicular-force pass from stationary bottlenecks. The central sampled minimum and the side maxima do not certify a two-saddle sequence or justify replacing the shared initial well with a new Pbcn endpoint. Its T-referenced expansion, F_xx=1.122022, is also a prospective branch-response diagnostic: the already registered substrate contains that direction. The existing G2 matrix must test whether that central branch persists, without adding a post hoc phase/path or an additional Hessian to the finite budget.

The terminal reversing centre at step 45 is Pa-3 at all three tolerances; its earlier step-32 Pbca/Pa-3/Pa-3 sensitivity remains recorded. Its amplitude in the selected rotated-T pattern triplet is essentially zero, but vanishing coordinates in a truncated representation do not identify a cubic structure. Its group label in this twelve-atom cell also does not identify a twenty-four-atom literature variant or a domain-wall motif. The PO-to-M peak retains its P1/P2_1 tolerance dependence rather than being standardized to a preferred label.

These are mechanism candidates for further relaxation, not discoveries of Pbcn-mediated switching or of an antipolar wall. A group symbol at one uniform snapshot does not map it onto the motif, boundary or energy of a published interface. Fixed and released lattice results must also remain separate rather than being pooled under a common phase label.

The preserving band's dominant residual changes from cell rows at step 14 to atoms at steps 25/27, then back to cell rows at steps 48 and 69. Its decreasing sampled maximum and rebound of the global force norm are compatible with a changing coupled configuration, not proof of a persistent single-block cause or of a particular optimizer's benefit. The preserving continuation ended normally on the unchanged force criterion; the reversing candidate subsequently passed as well. Whole-path polarization and transverse-stability audits remain distinct. Four ordinary passes alone neither complete G1 nor fix the stationary locations needed for local mode-strain predictions.

The forty terminal raw band tables are independently replayed with the audited 96-electron, 48-occupied-band contract. Minimum sampled indirect gaps are 4.2520, 4.1317, 4.5893 and 4.0173 eV for T/PO, M,

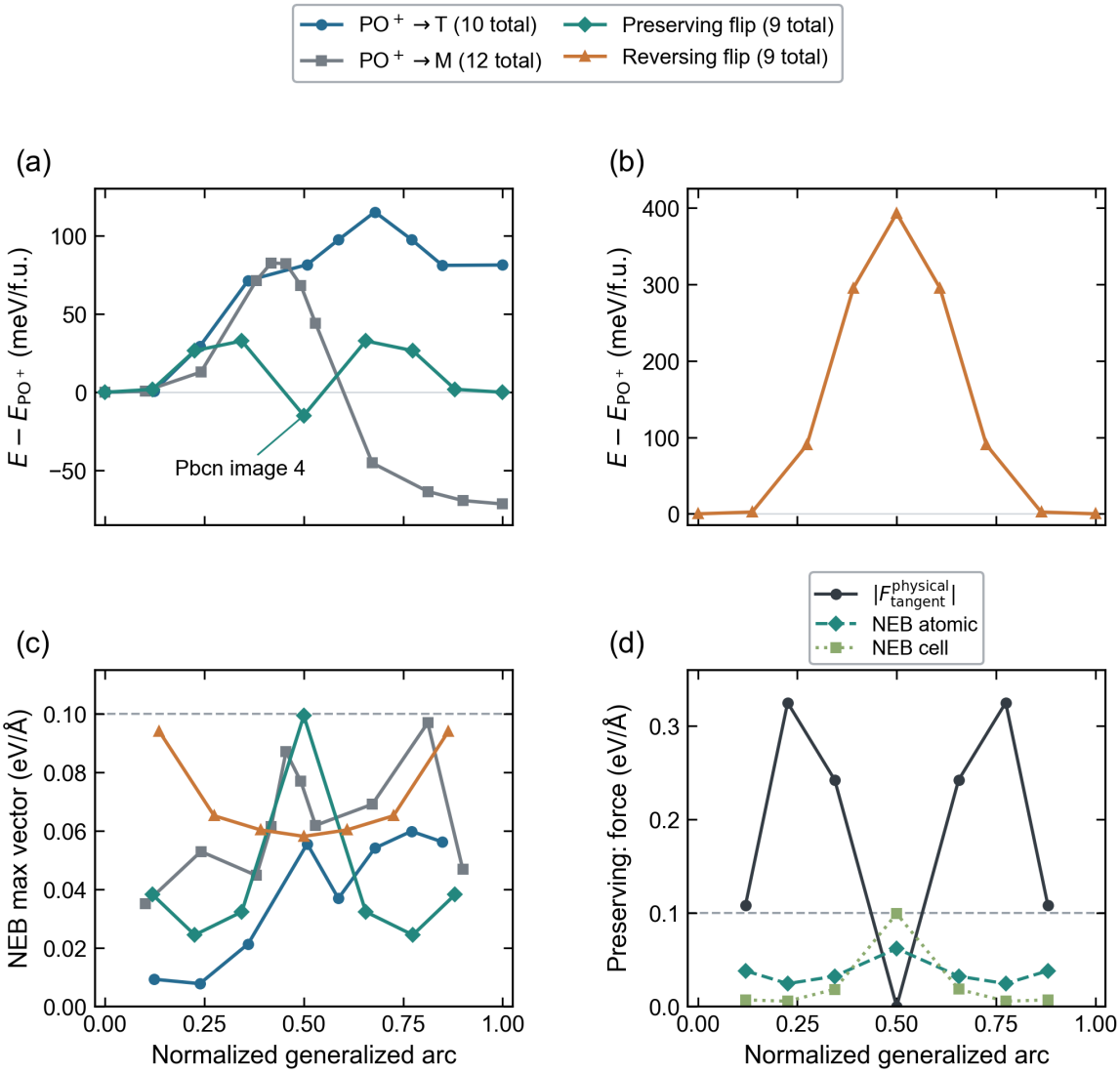


Figure A2. Complete same-PO+ terminal observations at steps 6/1/69/45, corresponding to Table A1. (a) Discrete low-energy profiles. (b) The higher, ordinary-converged reversing candidate is shown separately rather than omitted or clipped. (c) Ordinary NEB max-vector residuals on the moving images. (d) The preserving band's physical tangential-force magnitude and atomic/cell NEB residuals are different diagnostics in the same registered source metric. The dashed 0.10 line is the ordinary NEB target, not an independently established stationary-TS tolerance for the physical tangent. Lines connect calculated images only; endpoints have no NEB residual. No final energetic hierarchy or TS certification follows from this figure. The figure contract and CSV retain phase checks, original lifts and complete provenance. The earlier 6/39/69/32 figure and initial dated figure are retained with their original data and hashes, rather than overwritten.

preserving and reversing, respectively. These Gamma2x2x2 sampled gaps are not a full-Brillouin-zone insulating certificate or Berry-phase branch evidence. The separate native Berry protocol retains physical eR/V and spin-paired native $2eR/V$ periods; actual-cell reduced polarization, not a fixed Cartesian period, is needed along the changing lattice. Neither modular endpoint opposition nor a unique lift at finite sampled images establishes an absolute spontaneous polarization or excludes unsampled winding.

The missing R3 path measurements were then made at all fourteen existing switching interiors, reusing the three audited PO endpoint properties. Fourteen output-only SCFs reproduce baseline E/F/stress exactly at the raw-log precision; this is not a claim of zero total numerical error. Forty-two fixed-charge NSCFs use the registered $2 \times 2 \times 2$, $2 \times 2 \times 4$ and $2 \times 2 \times 8$ longitudinal quadratures. Maximum changes from the second to the third quadrature are 0.00063768 and 0.00083843 C/m^2 for the preserving and reversing candidates, below the predeclared 0.01 C/m^2 gate. No NSCF energy is used in a barrier.

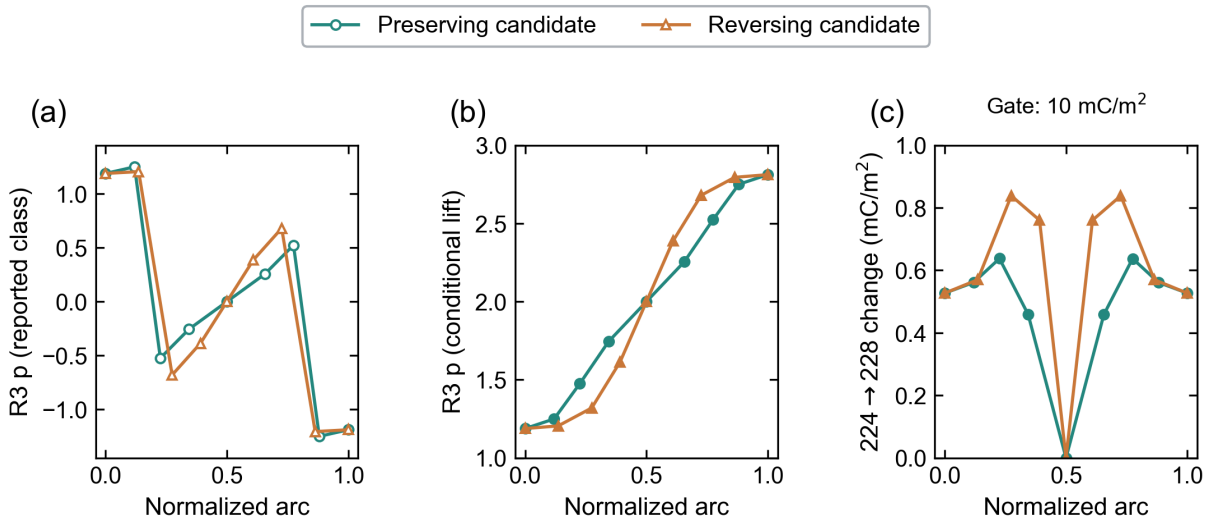


Figure A3. Native-path electronic property audit. (a) Reported classes in actual-cell reduced units $p = P_3/(e|R_3|/V)$. (b) Conditional nearest-sample lift retaining native period 2 and explicit initial integer gauge 0, not the smallest-absolute-P convention. (c) Changes from $2 \times 2 \times 4$ to $2 \times 2 \times 8$; the 10 mC/m^2 gate is stated above an axis showing the smaller measured range. Eighteen plotted rows comprise fourteen new interiors and cached endpoints, with common PO+ repeated. Connections join samples, not a continuous-path certificate. Source data and editable exports accompany the figure contract.

Both sampled lifts start at 1.188154, pass near 2 at their central nonpolar snapshots and end at 2.811846, giving the same reduced increment 1.623691. The minimum uncertainty-adjusted half-period margins are 0.7291 and 0.6120; no sampled link is ambiguous under the declared longitudinal sensitivity. These data do not justify calling the geometric candidates distinct polarization-winding sectors. The increment is not quantized, and a unique finite-sample lift still excludes neither unsampled winding nor transverse electronic components. It is not an absolute spontaneous-P selection.

Native integer h/min/s timing sums to 3155 s, or 28.0444 reported DFT core-hours at 32 ranks; the fourteen Slurm allocations sum to 4029 s, or 35.8133 core-hours including launch, copying and audit costs. The raw fourteen SCF and forty-two NSCF logs/band tables replay identically on HF and locally. Electronic settings and source geometries remain unchanged.

C.3. A residual pass does not resolve the sampled maximum

After the preserving band's ordinary pass, its cached physical forces and stress give positive-to-negative energy derivatives along the original fractional-coordinate/cell segments 2-to-3 and 5-to-6. Three registered

fractions, 0.25, 0.50 and 0.75, are calculated on each segment. All six SCFs converge with genuine 32-rank MPI and the original six electronic input hashes; the two sides are calculated rather than imposed as mirrors.

The new highest sample is 38.508448 meV/f.u., 5.702425 meV/f.u. above the original nine-image sampled maximum of 32.806023. Mirror-side maxima differ by only 0.0000174 meV/f.u.; that agreement is not a total barrier uncertainty. An inserted fifteen-image cached band still replays an ordinary maximum vector residual of 0.099372458 eV/Angstrom, without optimization or extra SCFs. Thus an unchanged residual pass can coexist with a materially different sampled maximum. The preserving curve in Table A1 and Figure A2 remains its original nine-image observation, not a sampling-resolved activation barrier.

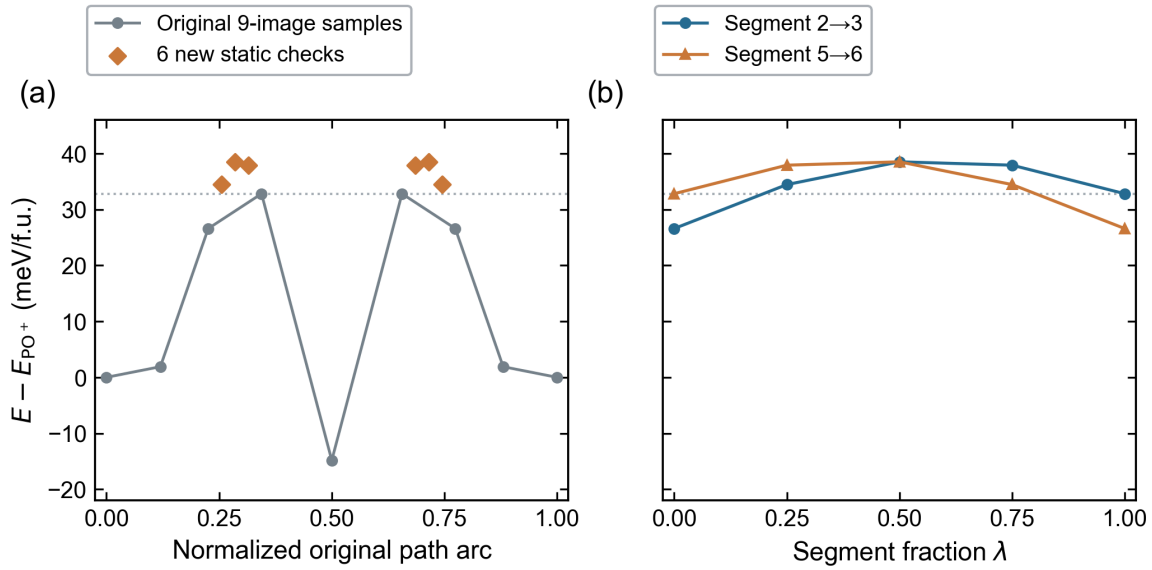


Figure A4. Six same-contract static checks of the preserving step-69 path. (a) Nine original samples and six added energies on the original normalized generalized arc. (b) The two tested segments, each with two original endpoints and three new static fractions. Straight connections are sample guides; the dotted line is the old sampled maximum. These are neither a smooth MEP fit nor a stationary-TS certificate. The source contains fifteen records, not fifteen new SCFs or independent replicates. Editable exports, all raw SCF audits and source data accompany the case.

The six SCF evaluations consume 542.98 seconds, or 4.8265 core-hours at 32 ranks. Allocation wall time is 601 seconds, or 5.3422 core-hours including launch and audit overhead. The checks validate a sampling concern chosen from already seen gradients; they are not the prospective material holdout. Their higher energies refer to the specified linear reconstruction, not an independently relaxed continuous path or a certified barrier error bound. Stationarity, transverse stability and matched-boundary predictions remain separate tests. The ordinary target, electronic inputs and no-CI policy are unchanged.

The same bounded screen was then applied to the other two already completed edges. Five additional static SCFs checked two near-peak fractions of T-to-PO and three fractions of PO-to-M segment3-to-4. The T-to-PO and PO-to-M sources, and the preserving flip, use an identical ordered periodic PO initial well; their shared energy reference is not an alignment of different endpoint wells. The M endpoint has

a maximum atomic force of 0.011905 eV/Angstrom and maximum stress component of 1.915095 kbar, within the registered endpoint screen.

Source path	Old peak	New static	Increase	Residual
T-to-PO	115.210164	115.221524	0.011360	0.060037
PO-to-M	71.582207	82.747071	11.164864	0.123408

Table A2. Five real reconstruction SCFs, not stationary transition states or an error-bounded continuous MEP. Energies are meV/HfO2 relative to common PO; residuals are eV/Angstrom. Each reconstructed band contains 12 total/10 moving images; only T-to-PO passes the ordinary 0.10 residual target. Force replay uses no optimization or further SCFs. The respective forward/reverse sampled barriers are 33.900331/115.221524 for T-to-PO and 82.747071/154.131671 for PO-to-M. Their differences equal the unchanged endpoint energy differences, -81.321193 and -71.384600 meV/HfO2. (Actual five-point audit)

Job 28380672 completed normally in 544 s at 32 CPUs. The five actual SCFs consumed 481.761 s, or 4.282322 core-hours; allocation cost was 4.835556 core-hours. All five retained the original six physical-input hashes, converged SCFs and complete E/F/stress. The PO-to-M screen estimated 81.718849 meV/HfO2; the actual sampled maximum of 82.747071 is a direct check, not independent validation of a barrier predictor. The increased residual required the bounded ordinary refinement of the twelve-image PO-to-M band described below. The prior nine-image source is preserved. No electronic retuning, CI or new mechanistic channel is introduced; neither a higher static sample nor a denser plot substitutes for transverse relaxation or stationarity.

The required M refinement subsequently passed at 0.0969640 eV/Angstrom after one new FIRE iteration from all twelve cached initial SCFs. Its sampled maximum is 82.679957 meV/f.u. and its fixed endpoints are unchanged. This is a continuation of the same edge and must not be called an acceleration benchmark: the source representation and force history differ from the earlier nine-image trajectory. Allocation cost is 9.884444 core-hours.

The reversing terminal band was checked at two registered near-peak fractions, 3-to-4 at 0.99 and 4-to-5 at 0.01. Both same-contract SCFs yield 392.822011 meV/f.u., below the existing 392.822905 maximum. The screen's tiny Hermite-predicted increase is therefore not an observed DFT increase or evidence of sub-microelectronvolt total precision. An eleven-image cached reconstruction replays 0.0940228 eV/Angstrom, without another optimization or SCF. These two measurements cost 1.309117 SCF core-hours and 1.813333 allocation core-hours; the finite peak-static budget has used thirteen of fourteen points, with no automatic use of the spare point. The final sampled values and their sampling/refinement histories are retained separately rather than pooled into stationary barrier estimates.

C.4. Reference choice changes the apparent compactness of the path

In the unweighted parent-scaffold chart, the three geometric T patterns capture 95.54% of the formation band's squared displacement at its sampled maximum, but only 31.43% at PO. Local compactness does not establish a global few-mode plane. Complete T-Gamma subspaces are evaluated separately in their mass metric; their displacement weights are not energy fractions.

The stronger published Cmma reference is registered without using modal coverage to choose its frame. Four tied registrations are retained. In the common T-origin mass metric, its four-direction span captures 27.18-27.19% at the formation maximum and 52.24% at PO, versus 89.69% and 30.42% for the three T patterns. The different reference and metric explain why those fractions must not be mixed with the preceding scaffold values. The lower or higher coverage of one chosen span is not a test of a nonlinear fixed-parent model. Published LDA directions are used as geometric candidates, not as our PBE Hessian or energy model. (Registration and scalar evidence)

C.5. A measured mixed response exposes a conditional-model limitation

Eight existing same-centre probes at a nonstationary historical image supply a restricted atomic-strain quadratic block. Releasing only its measured atomic direction softens the retained strain curvature by 12.36%/12.43% at the two amplitudes. Four previously seen longer-axis checks give a maximum energy residual of 0.04169 meV/cell; this is retrospective consistency, not an independent barrier forecast or a total uncertainty estimate.

The predicted atomic offset, 0.02165 Angstrom, places the specified release line outside the probe convex hull. The mixed response is measured, but a conditional DFT branch is unvalidated. No stationary T Hessian is spliced into this different centre, and stability of unmeasured directions is not assumed. This failed coverage test identifies what the local model cannot yet predict, rather than being concealed by a smooth contour. (Same-centre data and limits)

Appendix D. Matched endpoints and complete reference tables

D.1. Endpoint identities, screens and seen well responses

The qualified G1 candidate-connectivity review permits the registered mechanical comparison without declaring the pilot maxima certified saddles. The first G2 endpoint calculations fix the two substrate vectors to the relaxed T reference at nominal biaxial strain zero and release atomic coordinates and the third lattice vector, including tilt, at zero applied pressure. Zero strain therefore describes the T-referenced substrate, not free-cell conditions or zero strain relative to each phase's own lattice. All screened endpoints use the original ABACUS/PBE electronic contract.

The PO+ four-step canary retains $Pca2_1$ but reaches its step cap before the physical convergence screen. An exact-geometry continuation reuses only its terminal E/F/stress, with a fresh BFGS Hessian, and converges after two new SCFs. The registered T seed passes after its initial SCF, without an optimization step. Raw outputs reproduce both endpoint energies, forces and stress; the substrate vectors match. Symmetry labels agree at all three registered tolerances, without imposing symmetry on either calculation.

Endpoint	Maximum atomic force	Open traction norm	Relative energy
PO+	0.022285	1.928435	0.000000
T	0.000500	0.009595	12.359975
M	0.018207	0.843275	-92.540494
PO- preserving seed	0.014888	1.692852	-0.035259

Table A3. Screened endpoints at the common T-referenced, partially clamped zero-strain substrate. Forces are eV/Angstrom, open traction is kbar and energy is meV/HfO2 relative to the PO+ endpoint in this same ensemble. The endpoint targets are 0.03 eV/Angstrom and 2 kbar on open components; reaction stress in clamped components is not a failed convergence criterion. Neither the screen nor the phase label certifies a Hessian minimum. The PO- row names its registered seed; polarity and pattern registration remain separate from a space-group assignment. (Actual PO+ continuation; actual T endpoint and finite matrix)

The T-minus-PO+ well separation decreases from 81.321193 meV/HfO2 in the free-cell ensemble to 12.359975 meV/HfO2 under this substrate. The PO+ clamping energy cost is 68.961218 meV/HfO2, while the screened T reference is unchanged. This is an observed boundary-dependent well separation, not a smooth strain derivative between two points in one ensemble. In particular it determines neither the switching barrier nor the escape barrier: their maxima and possible intermediate branches still need matched path calculations. The M-derived endpoint also passes the same physical screen after seventeen BFGS steps and retains $P2_1/c$ at the three tolerances, without restoring its free-cell geometry or imposing its symmetry. Its energy is 92.540494 meV/HfO2 below PO+ in this same ensemble, which does not

determine the intervening escape maximum. The preserving opposite-polarity seed converges after seven BFGS steps and retains $Pca2_1$. Its observed -0.035259 meV/HfO2 offset is displayed without enforcing energy equality or interpreting it as an intrinsic polar bias or a numerical uncertainty bound. Electronic-polarity and ordered-pattern checks remain separate requirements before assigning the matched switching channels. The five registered zero-strain endpoint representations have now completed their physical screens.

The remaining reversing-minus endpoint independently passes the same screen after seven BFGS steps and eight SCFs. It does not introduce another physical phase. Composing the previously registered parent operations fixes a fractional translation (0.5,0.5,0) in the research frame and a species-preserving atom permutation. The two relaxed minus endpoints agree under this declared operation to 1.85×10^{-11} Angstrom, but differ by up to 1.36150 Angstrom at the original ordered indices under periodic boundaries. Their raw energies differ by 7.28×10^{-12} eV/cell; permuted forces and stresses covary to 3.0×10^{-10} eV/Angstrom and 2.38×10^{-12} eV/Angstrom^3, respectively. These are observed pair residuals, not a universal numerical precision estimate. The dominant geometric pattern has opposite signs in the fixed parent chart (-0.924790 and +0.924790 Angstrom). That sign is reference-origin dependent, not an intrinsic phase or electronic-polarization identifier. Endpoint translation equivalence does not establish whole-path equivalence: the original ordered correspondences and periodic lifts must be retained. The strict ordered cache correctly does not identify the two endpoints as the same input geometry. No production image is remapped by this read-only audit.

The two fresh PO+ continuation evaluations use 236.309 seconds, and the single T evaluation uses 125.001 seconds. The eight preserving-seed and eighteen M evaluations use 1960.598 and 2554.590 seconds, respectively. The eight reversing-seed evaluations use a further 1244.435 seconds. Together these thirty-seven fresh evaluations cost 54.408286 SCF core-hours at 32 ranks. The preceding five-SCF canary cost is recorded separately. Initial cached-result reuse avoids an identical SCF; it is not a benchmark of optimizer acceleration or evidence of the proposed predictive gain.

The first two registered +1% training endpoints also pass the same screens, each after six BFGS steps and seven fresh SCFs. PO+ retains $Pca2_1$; the T-derived endpoint instead has Ccce symmetry in the analyzed setting. This label is already present in its affinely strained seed before DFT: the substrate includes the original long axis and one short axis, so this partially clamped strain breaks the equivalence of the two original short axes. All seven evaluated geometries retain Ccce, and the atomic distortion remains in the original single geometric T-pattern direction, with amplitude 0.831575 to 0.885753 Angstrom and off-pattern residual below 1.33×10^{-14} Angstrom. It is therefore recorded as a strained-T descendant, without enforcing tetragonal symmetry or certifying a bulk phase minimum. Its well-energy separation from the same-substrate PO+ is 14.145883 meV/HfO2. These are seen training observations, not independent barrier predictions. The registered +1% M endpoint also passes after sixteen BFGS steps and seventeen SCFs, retaining $P2_1/c$; its force is 0.007755 eV/Angstrom and open traction is 1.452557 kbar. Its energy lies 114.746973 meV/HfO2 below PO+ in this same ensemble. This deeper well does not specify the escape barrier: the absolute path maxima, not endpoint depths alone, determine the proposed selectivity test. The finite matrix and raw records retain this distinction.

The two registered +1% minus endpoints have now independently passed the same screens, each after six BFGS steps and seven fresh SCFs. Both retain $Pca2_1$ at all three tolerances, with maximum atomic force 0.013275 eV/Angstrom and open traction 0.995807 kbar. Their fourteen evaluations cost a combined 14.250463 SCF core-hours at 32 ranks. These complete the ten endpoint representations, not ten distinct phases or certified local minima. Agreement between symmetry-related endpoint energies is not a numerical error bar for an intervening barrier.

The seen training endpoints already quantify a necessary well-response control. From zero to 1% substrate strain, PO+ rises by 6.806502 meV/HfO2, whereas the screened M representation falls by 15.399977 meV/HfO2. The M-minus-PO+ well separation therefore changes by -22.206479 meV/HfO2. This is not a measured change in the M escape barrier. The two preregistered B1 assumptions make this distinction explicit: fixing the absolute bottleneck energy gives the same -6.806502 meV/HfO2 barrier-response

increment for every channel; allowing it to follow the final well gives the channel-dependent increments in Table 2. No anchor barrier, unseen condition or new mechanism is inferred from these increments.

A calculator-independent work contraction evaluates the local imposed-plane partial as $V\sigma : [H_\varepsilon^T H^{-T}]$, with ASE row cells, tensile stress and $H_\varepsilon^{0:2} = H^{0:2}/(1+\varepsilon)$ while the open third vector is held fixed for this partial. It is not an exact relaxed-branch envelope derivative at the nonzero screened force and open-traction residuals. Endpoint stress-work trapezoids differ from the observed well-energy changes by 0.019–0.347 meV/HfO2. Including atomic and released-cell residual work along the ordered endpoint configuration chord gives defects of -0.001–0.204 meV/HfO2. These two-endpoint diagnostics mix unmeasured curvature and finite-relaxation/numerical effects; they are not independent sampling-error bounds or proof of a smooth stationary branch. The raw-work analysis retains the signed defects, exact sources and all null increments. Actual path-maxima response must be compared with these well-only controls before attributing an improvement to a new mode mechanism or prediction method.

D.2. Complete sampled forward/reverse values

Table A4. Eight ordinary-converged training bands. Energies are meV/HfO2 under the same P=0 electronic contract; reverse peaks and well differences retain the endpoint-inclusive convention. These are not certified TS values.

Candidate	Forward discrete peak	Reverse discrete peak	Final-minus-initial well energy
PO-to-M, common T-plane clamping	98.458122	190.998616	-92.540494
PO-to-M, +1% substrate strain	97.227719	211.974692	-114.746973
PO-to-T, common T-plane clamping	84.416624	72.056649	12.359975
Reversing-pattern switching, same clamping	338.968965	339.004224	-0.035259
PO-to-T, +1% substrate strain	96.763650	82.617767	14.145883
Preserving-pattern switching, same clamping	96.828183	96.863442	-0.035259
Preserving-pattern switching, +1% substrate strain	125.424431	125.424431	-0.000000
Reversing-pattern switching, +1% substrate strain	294.944634	294.944634	0.000000

D.3. Complete atomic reference comparison

Table A5. Squared atomic-displacement and atomic-gradient coverage (%) at each sampled peak in the original T chart. The denominators differ; cell response remains separate. All four Cmma frames 103/106/114/127 are retained, and their ranges are not uncertainty intervals or model errors.

Candidate	Atomic reference space	Rank	Peak displacement	Peak gradient
PO-to-T	T geometric triplet	3	99.0957	14.3013
PO-to-T	Two lowest complete T optical doublets	4	0.2411	20.3963
PO-to-T	Registered Cmma span	4	16.6635–16.6636	54.0718–54.0765
PO-to-T, +1%	T geometric triplet	3	99.0045	69.7245
PO-to-T, +1%	Two lowest complete T optical doublets	4	0.2822	2.0694
PO-to-T, +1%	Registered Cmma span	4	18.2383–18.2386	10.5750–10.5766
PO-to-M	T geometric triplet	3	57.9237	9.0282
PO-to-M	Two lowest complete T optical doublets	4	12.0962	32.7096
PO-to-M	Registered Cmma span	4	36.4746–36.4865	93.5085–93.5087
PO-to-M, +1%	T geometric triplet	3	54.9251	3.9008
PO-to-M, +1%	Two lowest complete T optical doublets	4	12.9756	31.7793
PO-to-M, +1%	Registered Cmma span	4	38.4024–38.4141	91.5873–91.5896
Reversing-pattern switching	T geometric triplet	3	26.2915	0.0320
Reversing-pattern switching	Two lowest complete T optical doublets	4	16.2196	10.0215
Reversing-pattern switching	Registered Cmma span	4	46.5637–46.5639	65.1046
Preserving-pattern switching	T geometric triplet	3	82.6301	12.9583
Preserving-pattern switching	Two lowest complete T optical doublets	4	<0.0001	<0.0001
Preserving-pattern switching	Registered Cmma span	4	14.8336–14.8337	52.7203
Preserving-pattern switching, +1%	T geometric triplet	3	52.1751	32.8790
Preserving-pattern switching, +1%	Two lowest complete T optical doublets	4	<0.0001	<0.0001
Preserving-pattern switching, +1%	Registered Cmma span	4	16.9707–16.9708	49.4667
Reversing-pattern switching, +1%	T geometric triplet	3	16.1410	<0.0001
Reversing-pattern switching, +1%	Two lowest complete T optical doublets	4	18.1850	22.4841
Reversing-pattern switching, +1%	Registered Cmma span	4	52.8497–52.8499	63.5220

The ordered cyclic G2 frame change is inverted on geometry-only copies before transport to the original T chart. Native E/F/stress and ordinary residuals are replayed in their production frame. Dual gradients use $g_q = -F[H_T^{-1}H]^T/\sqrt{m}$, with the same translation-free gauge for the gradient and each declared reference span. The source translation-overlap norm 0.0036854 is retained; no ASR repair is made to modes or raw forces. Published LDA Cmma vectors are geometric controls, not our PBE curvature.

D.4. Complete local joint-curvature measurements

Table A6. Smallest internal eigenvalues of the symmetrized finite-step matrices, in eV/Angstrom squared. Steps are in Angstrom. The last column counts negative eigenvalues after excluding the internal path tangent. Counts are raw signs, not uncertainty-resolved stationary indices. The prescribed substrate control and three rigid translations are excluded from these internal spectra. T and preserving denote the original sampled zero-strain peak centers, not certified TSs.

Center	Step	First	Second	Third	Internal negative	Tangent-complement negative
T	0.01	-0.276193	-0.164648	0.053216	2	1
T	0.02	-0.273596	-0.161974	0.050132	2	1
Preserving	0.01	-1.271819	-0.447450	0.173653	2	1
Preserving	0.02	-1.281821	-0.455584	0.172180	2	1

Full native responses, gradients, reciprocity and two-step diagnostics are retained in the complete local-curvature case. No ASR or source-force adjustment is applied. A positive numerical complement at a nonstationary center is not post-relaxation stability.

D.5. Strong references in the joint released space

Table A7. Minimum released eigenvalue in eV/Angstrom squared, followed by its raw negative count in parentheses. Columns distinguish both centers and finite steps. All four Cmma registrations are retained, including numerically tied rows. Each reference is an atomic span in the common 36-dimensional internal problem; cell directions remain released. These unequal retained ranks are not matched-size competing predictors. The complete T optical reference is a strong control, not an excluded failure.

Atomic reference	Retained rank	Released dimension	T, 0.01	T, 0.02	Preserving, 0.01	Preserving, 0.02
T geometric triplet	3	33	-0.230562 (2)	-0.227205 (2)	-1.271818 (2)	-1.281820 (2)
T complete optical doublets	4	32	-0.175388 (1)	-0.173389 (1)	-0.246013 (1)	-0.253469 (1)
All T optical	33	3	2.956214 (0)	2.956557 (0)	3.090798 (0)	3.092172 (0)
Cmma 103	4	32	-0.153434 (1)	-0.151224 (1)	-0.447452 (2)	-0.455596 (2)
Cmma 106	4	32	-0.153437 (1)	-0.151227 (1)	-0.447451 (2)	-0.455594 (2)
Cmma 114	4	32	-0.153437 (1)	-0.151227 (1)	-0.447451 (2)	-0.455594 (2)
Cmma 127	4	32	-0.153434 (1)	-0.151224 (1)	-0.447452 (2)	-0.455596 (2)

The reference-complement records retain separate atomic-only and joint release tests. This table uses the joint blocks; published reference frequencies are not substituted for PBE curvatures. Raw signs and registration spread do not bound physical error. The all-optical retained Schur block has two raw negative directions in each case despite its positive released block.

D.6. Bounded fixed-coordinate relaxation comparison

Table A8. Actual new SCFs, final released-gradient norms (eV/Angstrom) and native call seconds at fixed retained coordinates. A dash denotes no gradient-converged final candidate.

Center	Initialization	New SCFs	Final released norm	Native call seconds
T	Identity, budget exhausted	12	-	1489.02
T	Curvature seed	2	0.02630	244.81
Preserving	Identity, budget exhausted	12	-	1118.94
Preserving	Curvature seed	5	0.01959	452.40

Each arm has the same twelve-attempt limit and 0.03-eV/Angstrom released-gradient norm target. The last evaluated trial is not substituted for a missing final candidate. Native seconds exclude cached-center evaluation and the preceding 296 curvature SCFs; earlier path costs are also additional. Identity costs are censored, so ratios of these call counts are not measured speedup factors.

The native canary records retain all 31 new SCFs, attempt reservations, allocation times and cache-only optimizer replays. Final candidate energies are -9782.63681659 eV/cell (T) and -9782.63600314 eV/cell (preserving), using four-formula-unit cells. No unseen-strain label, new full path or post-relaxation curvature is inferred from this comparison.

D.7. Eligibility of restricted references at shared observables

The geometry-eligible counts use the original guard and a denominator of 24 planned nonzero coordinates per center. Counts are the same at both probe steps. The atomic reference rank is distinct from the allowed dimension, which includes three open-cell directions. The coupled full-space arm retains two coordinates

and releases 34; it is not a nominal atomic reference of rank two. Outside-reference dimensions are clamped in the restricted arms, not silently eliminated.

Table A9. Geometry-eligible seen nonzero coordinates under the original guard.

Reference	Atomic rank	Allowed dimension	T eligible	Preserving eligible
T geometric triplet	3	6	8/24	0/24
T complete optical doublets	4	7	10/24	8/24
All T optical	33	36	24/24	16/24
Cmma 103	4	7	6/24	4/24
Cmma 106	4	7	6/24	4/24
Cmma 114	4	7	6/24	4/24
Cmma 127	4	7	6/24	4/24
Coupled full-space	-	36	24/24	16/24

There is no nonzero coordinate at which all arms have both a candidate label and an eligible model geometry. Common-set error statistics are therefore absent, not zero. No reference is selected by its own-subset error, and the four Cmma frames are not pooled into a larger basis. The complete scalar records retain all point identities, signs, origin biases and eligibility decisions.

D.8. Complete matched atomic-only treatment

Table A10. Origin-aligned maximum model discrepancies on the same four axial coordinates per center, in meV per twelve-atom cell. F freezes omitted coordinates, A releases atomic coordinates at the frozen-Q seed cell, and J jointly releases atomic and permitted cell coordinates. Probe steps are Angstrom. The comparison is on seen training labels, not independent errors.

Center	Probe step	Frozen	Atomic-only	Joint
T	0.01	0.416264	0.501879	1.327177
T	0.02	0.403160	0.515704	1.346071
Preserving	0.01	0.075880	4.349824	4.671404
Preserving	0.02	0.116535	4.399770	4.720758

All ten atomic-only candidates and 36 new SCFs passed the complete native audit, including unchanged electronic inputs and exact optimizer replay. The original twenty model-point predictions and both probe steps were retained without refitting. The measured atomic-release call proxy was 39.749558 core-hours; the allocation used 40.684444 core-hours. The eight frozen statics, original 296 curvature SCFs and all previous joint-path and candidate preparation remain additional. The unused part of an attempt ceiling is not a speedup measurement. The complete matched records retain signed absolute and origin-aligned treatment differences. These candidate differences are not uniquely separable mode energies or evidence of a stable conditional surface; post-release curvature and independent predictions remain separate requirements.

Appendix E. Detailed response and model-validity diagnostics

These analyses reuse the declared training paths and local candidates. They support the mechanical and representation checks in the main text, without adding independent observations or changing the electronic contract.

E.1. Endpoint closure can conceal a bottleneck-region discrepancy

The signed-work diagnostic uses the eight completed clamped training bands, not earlier unfinished snapshots or the reserved strain condition. All nine raw E/F/stress frames per band are reparsed, with the same ordered continuous lift, original T mass chart, permitted open-cell space and electronic contract. These 72 frame records contain 66 distinct raw-log/geometry identities: four zero-strain paths share one initial PO+ cache, and the four +1% paths share another. They are neither new SCFs nor independent statistical replicates. Every energy change is relative to that band's own matched PO+ well.

Table A11 compares raw sampled peak energies with the full endpoint-gradient trapezoid accumulated to the same image. Peak and endpoint residuals use the same sign, raw energy change minus accumulated work. Units are meV/HfO2 throughout; the Hf4O8-cell values are multiplied by 250.

Training band	Raw peak	Work to peak	Residual at peak	Residual at endpoint
PO-to-M, zero strain	98.458122	77.560379	20.897743	3.006016
PO-to-T, zero strain	84.416624	75.955937	8.460687	0.728018
Reversing flip, zero strain	338.968965	303.693643	35.275323	0.115373
PO-to-T, +1% strain	96.763650	87.345393	9.418257	1.997185
Preserving flip, zero strain	96.828183	82.304235	14.523948	0.032398
PO-to-M, +1% strain	97.227719	75.922579	21.305139	2.658882
Preserving flip, +1% strain	125.424431	111.393911	14.030520	0.000050
Reversing flip, +1% strain	294.944634	259.087641	35.856993	-0.000001

The reversing candidate illustrates the distinction most clearly. Its endpoint closure is only 0.115373 meV/HfO2, whereas the discrepancy at the highest sample is 35.275323. Small final closure therefore does not certify energy-resolved consistency through the transition region. Conversely, the present nine-sample trapezoid cannot determine whether this discrepancy comes from coarse quadrature, physical force/stress error or both. It is not a measured barrier uncertainty and does not justify retuning the electronic inputs or declaring an ordinary-converged band failed.

The coordinate pullback itself passes a distinct implementation check: direct native Cartesian-force/ASE-stress contractions agree with the dual atomic/open-cell work within 1.12×10^{-16} eV per cell. This algebraic agreement does not test DFT accuracy. Under the T geometric triplet, signed retained work to the respective peaks is -0.111271, 2.666921, 195.447066, 1.367408, -0.165926, 0.011623, 3.753494 and 147.404416 meV/HfO2, in Table A11 order. Omitted atomic work is 68.756917, 72.789643, 105.539695, 85.773095, 82.500795, 67.611606, 107.167048 and 108.609913; open-cell work is 8.914733, 0.499373, 2.706882, 0.204891, -0.030633, 8.299349, 0.473369 and 3.073313. The explicit rigid translation terms are also retained, at most 4.63×10^{-9} in these units at the displayed peaks, rather than removed by a source ASR repair. These signed terms sum to the full sampled work, not to a unique energetic allocation among modes.

Strong references remain in the comparison. The two complete T optical doublets yield retained work 12.359091, 11.900861, 0.782574, 11.007299, 18.977823, 11.650637, 26.369936 and 6.179305 in the same order. Across all four registered Cmma frames, retained-work ranges are 55.618955 to 55.619265 for zero-strain M; 62.066996 to 62.068212 for zero-strain T; 63.846564 to 63.846572 for reversing; 67.984207 to 67.985028 for +1% T; 47.684643 to 47.684652 for preserving; 55.402398 to 55.403003 for +1% M; 26.439087 to 26.439089 for +1% preserving; and 23.010446 to 23.010450 for +1% reversing. Each representation includes its full omitted atomic contribution and the same open-cell work; ranks three and four are not treated as identical-dimensional models. This reference dependence is not a conditional-energy or forecast-accuracy ranking. In particular, the reversing triplet's integrated work and its small instantaneous peak-gradient fraction (Table A5) answer different questions and need not track one another.

The four original scalar reports, preserving report, +1% M report and eight-band raw-frame replay and cohort check thus add an energy-gradient consistency test to geometric coverage, without selecting G3 bottlenecks, a reference winner or unseen labels. Measured conditional response, stability and the prospectively frozen strong-control comparison remain necessary for a predictive claim. This diagnostic uses existing data only and does not open an additional sampling or Hessian budget.

E.2. Cross-strain work separates imposed control from configuration change

The second work diagnostic pairs equal image indices between the two training conditions in the single controlled-coordinate chart of Appendix G.4. All four pairs retained their sampled peak index (M:3; T and both flips:4). The matched-peak energy changes therefore reproduce Table 1 exactly, while the signed terms in Table A12 partition the declared endpoint trapezoid.

Table A12. Cross-strain peak-relative work in meV/HfO2. Terms are relative to the same initial-pair work; the final column is energy change minus full trapezoidal work, not a barrier-error estimate. The retained rigid-translation terms are smaller than 2.411e-10 at these peaks.

Candidate	Observed peak change	Atomic work	Open-row work	Substrate work	Energy minus work
PO-to-M	-1.230403	2.720687	-0.174328	-3.835247	0.058485
PO-to-T	12.347027	-0.228947	0.220068	12.275095	0.080811
Preserving flip	28.596248	36.895288	-0.078916	-12.858888	4.638764
Reversing flip	-44.024331	-32.332083	-0.270001	0.072814	-11.495061

In this connection, the T candidate's work was dominated by the prescribed substrate term. For preserving, that term had the opposite sign from the observed peak change: +36.895288 atomic work and a +4.638764 residual accompanied the -12.858888 substrate term. The isolated mechanical partial therefore cannot replace the observed finite path response. It is not a measurement that a frozen strain slice would lower the barrier: the endpoint gradients here belong to two different internally optimized configurations.

The reversing residual was -11.495061 meV/HfO2. It is preserved rather than attributed solely to DFT noise or a particular coupling. No intermediate cross-strain geometries were calculated. Independent Cartesian force/stress and dual-chart contractions agreed within 6.662e-16 eV/cell, an algebraic implementation check rather than physical accuracy evidence. The 36-pair source analysis reuses the same 72 frame records/66 distinct identities, without new SCFs, held-out inputs or reference selection. The next inference requires measured local curvatures, stable conditional branches and frozen independent tests.

E.3. The extra negative direction is already present in the atomic sector

The additional raw negative direction at each sampled peak is not created solely by opening the cell. Both atomic restrictions already had two negative eigenvalues at both finite steps, as did the joint matrices; their open-cell restrictions had three positive eigenvalues (Table A13). For these measured quadratics, coupling changed the negative curvatures without adding another negative direction. This observation is specific to the two nonstationary centers, not a general rule excluding coupling-induced instabilities or an uncertainty-resolved stationary index.

Steps are in Angstrom and the three curvature columns in eV/Angstrom squared. The last column is the dimensionless open-cell trace of the two-dimensional negative subspace, not a percentage of a barrier energy. The same length scale L=5.12968067458423 Angstrom is used in every row.

Table A13. Atomic and cell restrictions of the original sampled-peak matrices.

Center	Step	Minimum atomic	Minimum joint	Minimum cell	Cell trace
T	0.01	-0.240169	-0.276193	2.956214	0.0205773
T	0.02	-0.237806	-0.273596	2.956557	0.0206105
Preserving	0.01	-1.271818	-1.271819	3.090798	1.04022e-06
Preserving	0.02	-1.281815	-1.281821	3.092172	3.09248e-06

The T negative subspace contained more open-cell admixture than the preserving subspace in this metric. However, the complete atomic-cell coupling operator norms were approximately 7.72 and 8.10 eV/Angstrom squared, respectively, at the smaller step. A large total coupling norm therefore did not identify a strongly cell-mixed unstable direction. Conversely, small open-cell weight at one sampled configuration cannot establish insensitivity of a complete barrier to the prescribed substrate. The substrate response and the permitted open-cell fluctuations are different quantities. All four restrictions and the nonzero-gradient formal reductions are retained in the atomic/open-cell analysis.

E.4. A useful curvature seed is not a finite-domain energy predictor

The bounded optimizer result in Appendix E.6 did not establish the accuracy of the original quadratic over finite retained displacements. In the first-step eigenplane, the retained/released coupling was numerically zero by construction; its Schur curvature correction was of order 10^{-28} eV/Angstrom squared in norm. Formal release changed the energy offset but supplied no appreciable new shape correction in that plane. This algebraic null is not a prediction improvement.

After origin alignment, the original quadratics still disagreed with the six common-coordinate DFT candidates (Table A14). Both probe steps are reported, not selected according to agreement. The maximum absolute shape discrepancies were 8.52–8.59 meV/cell for T and 20.52–20.72 meV/cell for preserving, each at $q = (-0.2, 0)$ Angstrom. The corresponding signed errors were positive for T and negative for preserving. All nine T candidates and all six preserving candidates in the audited source snapshot remain in the records; the common cohort is used only to make the table's coordinate coverage explicit.

All three energy columns use meV per twelve-atom cell, not meV/HfO2. Origin release is relative to each original center; the shape discrepancy uses each model's and candidate's own seen origin. Probe steps are in Angstrom. These are seen-training diagnostics, not independent errors or continuous-path barrier errors.

Table A14. Unfitted original quadratic versus released DFT candidates.

Center	Probe step	DFT origin release	Model origin release	Max. shape discrepancy
T	0.01	-0.652142	-0.866958	8.524106
T	0.02	-0.652142	-0.874429	8.587800
Preserving	0.01	-49.484930	-54.521222	20.521839
Preserving	0.02	-49.484930	-54.705828	20.720640

The candidate-energy axial second differences at 0.1 Angstrom were (-0.405128, -0.243087) eV/Angstrom squared for T and (-0.337763, -0.184576) for preserving. They are finite differences of candidate energies, not infinitesimal Hessians of certified conditional minima. Nonlinear response, finite-difference effects, branch changes and residual relaxation can all contribute to the model disagreement; these observations do not distinguish their individual contributions. Nor does their size define a physical uncertainty bound. The quadratic comparison keeps the affine released gradient and both original probe steps.

These results separate two uses of local curvature: an initialization that is updated during optimization, and a fixed finite-domain energy model. The former can assist local relaxation even when the latter misses

the observed shape. A matched frozen-coordinate comparison is needed before assigning the discrepancy to the relaxation response. Eight such static controls, four per center, have now completed under the unchanged electronic contract. Native energy, force, stress, input and 32-MPI checks passed for all eight. They use the original cached origins and seen released counterparts, reserve training slots only, and do not replace the independent tests.

The subsequently completed T candidate sheet exposed a separate finite-convergence issue. Its 25 points required 118 new SCFs in addition to the cached released origin. Final released-gradient norms ranged from 0.00653 to 0.02978 eV/Angstrom, and the largest total internal displacement was 0.46378 Angstrom within the original 0.5-Angstrom domain. Sampled energies spanned -38.531 to 0 meV/cell relative to that released origin. These are fixed-coordinate candidates, not an end-to-end MEP or certified stable conditional minima.

Across the 40 nearest grid edges, the largest absolute discrepancy using only retained endpoint work was 1.209 meV/cell. On that edge the signed released term was 1.094 meV/cell, leaving a full quadrature discrepancy of 0.115 meV/cell. Over all edges the largest absolute full discrepancy was 1.039 meV/cell. Across 16 elementary closed loops, the largest absolute retained and full endpoint circulations were 2.000 and 0.578 meV/cell, respectively; these maxima occurred on different loops. This complete-sheet diagnostic retains finite released residuals and unsampled quadrature separately. The values are not DFT, interpolation or barrier-error bounds and do not certify smoothness or stability. They also do not identify the cause of the original quadratic's finite-domain disagreement.

The completed preserving batch did not attain the same coordinate coverage. All 25 coordinates were processed, but only 17, including the cached origin, reached the common released-gradient target. The eight remaining coordinates stayed as holes. All 172 new SCFs passed native energy, force, stress, unchanged-input and genuine 32-MPI checks, and all 24 nonzero-coordinate optimization trajectories replayed exactly. Seven holes exhausted twelve evaluations; one stopped after ten without reaching the target. They consumed 94 of the 172 SCFs. The batch required 193.859 completed-call proxy core-hours and 195.120 allocation core-hours, excluding the original curvature and cached preparation.

Cache-only replay attributed every recorded geometry rejection in these eight trajectories to the original total-displacement radius, not to a changed electronic input or a failed SCF. Their last completed trials lay at radii 0.49971–0.50000 Angstrom within the registered 0.5-Angstrom domain; none of the completed trials met the 0.03 eV/Angstrom released-gradient target. This establishes contact with the optimizer's domain boundary, not nonexistence of a solution, boundary stationarity or termination of a physical branch. The complete audit and hole replay retain the original decisions and stopping reasons. We do not fill these holes by interpolation or equate batch completion with a stable surface.

The matched controls locate the larger preserving-channel discrepancy in the release response, within this four-coordinate training cohort (Table A15). Frozen shape discrepancies stayed below 0.12 meV/cell for both original probe steps, while released discrepancies reached 4.67–4.72 meV/cell. The DFT release change relative to its origin reached 4.607 meV/cell. At $q = (-0.1, 0)$ Angstrom for the first-step model, the signed frozen discrepancy -0.064445 and missed release-shape term -4.606959 summed to the released discrepancy -4.671404 meV/cell.

All energy columns are meV per twelve-atom cell; probe steps are Angstrom. Separate column maxima may occur at different coordinates and are not subtracted to form a decomposition. All 16 model-point records are retained in the matched ablation. These are training-only diagnostics, not independent predictions or physical uncertainty bounds.

Table A15. Matched frozen/released shape discrepancies on four axial coordinates per original center.

Center	Probe step	Max. frozen shape	Max. released shape	Max. DFT release change
Preserving	0.01	0.075880	4.671404	4.606959
Preserving	0.02	0.116535	4.720758	4.606959
T	0.01	0.416264	1.327177	1.168761
T	0.02	0.403160	1.346071	1.168761

Signed cancellation also matters. At T $q = (0.1, 0)$ Angstrom, the first-step frozen discrepancy -0.416264 and release-shape term +0.378430 partly cancelled, leaving -0.037833 meV/cell. This near agreement at one coordinate did not establish general model accuracy. The ablation distinguishes frozen-shape mismatch from missed release-shape change, but does not isolate anharmonicity, finite-step effects, finite released residuals or branch changes as a unique microscopic cause. Nor does it replace the full T/Cmma strong-control or independent-prediction matrix. The eight statics used 8.137 completed-call proxy core-hours and 9.076 allocation core-hours; curvature and prior candidate preparation remain additional.

E.5. A complete reference removes a basis advantage, not a finite-domain error

The complete T optical reference and the coupled-coordinate representation gave the same constrained quadratic, not competing accuracy results. Using the same measured H, affine g, fixed Q and mechanical space, the 33 optical directions plus three open-cell directions spanned all 36 internal coordinates. Across both centers, both probe steps and all 25 registered Q values, the largest energy difference between the constrained reference solve and Schur solve was 2.64×10^{-16} eV/cell; the largest coordinate difference was 1.56×10^{-14} Angstrom. These differences measure numerical realization of an algebraic identity, not DFT accuracy.

Both representations retained the same finite-domain disagreement with the completed DFT candidates (Table A16). The larger T maximum compared with Table A14 reflects its expanded seen-coordinate cohort, not a newly observed independent error. The preserving holes were excluded as missing labels, not replaced with their last optimization trials. No coefficients were fitted, and subtracting each seen origin remained a retrospective shape diagnostic. Counts exclude the origin and share the denominator of 24 planned nonzero coordinates. All listed candidates passed the original geometry guard for both models; probe steps are in Angstrom.

Table A16. Shared full-space shape discrepancies (meV per twelve-atom cell).

Center	Probe step	Seen nonzero candidates	Max. shape discrepancy
T	0.01	24/24	28.776664
T	0.02	24/24	28.872042
Preserving	0.01	16/24	20.521839
Preserving	0.02	16/24	20.720640

The restricted references did not provide a common nonzero comparison cohort in these fixed observables. Their individual eligible sets differed (Appendix D.7), and the all-reference intersection was empty at both probe steps. Thus neither extrapolated values nor errors computed on different subsets were used to rank them. This is a compatibility limitation of the selected spans, observables and finite domain, not evidence that a native Cmma-coordinate energy model fails. The shared-observable control retains all seven registered references, four Cmma frames and the full-space control: 800 algebraic model-point records, with zero new DFT calls. The original 296 curvature SCFs, 118/172 conditional-sheet SCFs and all other preparation remain additional; equal preparation cost or speedup was not established. These comparisons delimit the claimed contribution without replacing a prospective native-chart or independent material test.

E.6. Local curvature, conditional release and bounded optimizer evidence

Direct joint-curvature measurements reveal an additional limitation of a one-coordinate description at the two sampled peaks. At both finite steps, each internal matrix has two negative eigenvalues; removing the internal path tangent still leaves one negative direction (Table A6). For T escape, the lowest two values are approximately -0.276 and -0.165 eV/Angstrom squared at the smaller step. For preserving switching they are -1.272 and -0.447. These are nonstationary, finite-step curvatures in a joint atomic-cell length metric, not phonon frequencies or certified saddle indices. For these measured matrices, retaining only one internal direction cannot leave a positive released complement. This is a property of the measured quadratic, not a new reduction theorem.

The registered reference spaces must be tested against this same internal problem rather than displacement coverage alone. The rank-three T triplet, rank-four optical control and each of the four rank-four Cmma registrations leave negative released curvature at both centers and steps (Table A7). However, retaining all 33 T optical directions leaves a positive three-cell- direction complement, while its retained Schur block still has two negative directions. Thus the evidence does not show that all fixed references fail. The ranks and released dimensions differ, and a complete quadratic response remains invariant under valid basis partitions as described in Appendix G.2.

Releasing the remaining coordinates at fixed two-dimensional coordinates lowers the T-center energy by 0.00065214 eV per twelve-atom cell and the preserving-center energy by 0.04948493 eV per cell. The curvature-seeded arms reach released-gradient norms 0.02630 and 0.01959 eV/Angstrom in two and five new SCFs, respectively; neither identity arm reaches the common target within twelve attempts (Table A8). This is a bounded local result, not a whole-NEB acceleration benchmark. The preceding 296 curvature SCFs and earlier path costs are additional, so no net cost advantage follows from these call counts alone. Post-relaxation released stability had not been measured in that bounded canary; the later checks follow below.

The two resulting candidate energies differ by only 0.00081345 eV per cell, but energy proximity does not establish a common mechanism or branch. They retain a 0.19178-Angstrom translation-free distance in the common ordered T chart, without atom rematching or rotation. Moreover, each retained plane is anchored at its own original center. These local fixed-coordinate results therefore do not replace any sampled G2 barrier, establish branch merger or certify a transition structure. They motivate checking conditional equilibration and branch stability before using a low-dimensional surface for prediction.

E.7. A shallow omitted direction after conditional release

Complete post-release stencils comprised 136 native SCFs per origin, including both 0.01/0.02-Angstrom steps and all 34 released directions. At T the minimum symmetric-raw eigenvalues were 0.164065/0.157596 eV/Angstrom squared; at preserving they were -0.022077/-0.024090, each with one negative direction. Released-gradient norms were 0.026300/0.019591 eV/Angstrom. Both nonsymmetric blocks and their step disagreement remain available, not repaired into positive matrices or total physical-error bounds. The centers are local path-associated candidates, not endpoint phases or TSs.

The preserving soft vectors had squared overlap 0.999997889 between steps. Their oxygen/open-cell squared weights were 0.999990/0.000009 in the original length metric, not energy fractions or mass-weighted phonon labels. The 31-dimensional fixed-cell restriction retained minima -0.022058/-0.024070 eV/Angstrom squared. Formally retaining the first-step soft direction beside the unchanged two coordinates left raw positive 33-dimensional blocks at the same point, with minima 0.384878/0.375093. This motivates an explicit-coordinate test; it does not supply the new full retained Hessian or a stable nonlinear three-coordinate surface.

All four finite-amplitude statics lowered the reported energy relative to the same cached origin (Table A17). Changes are per twelve-atom cell, four formula units; gradients follow the fixed unit direction in the

length chart. Case field t_A denotes the paper's internal amplitude ζ, not the external strain control t. Full native sources, energy/gradient/stress and both frozen quadratic predictions remain in the four-point case.

Table A17. Four frozen directional training statics at the released preserving origin.

Amplitude (Angstrom)	Energy change (meV/cell)	Directional gradient (eV/Angstrom)
+0.05	-0.050533	-0.000841328
-0.05	-0.050260	+0.000840951
+0.10	-0.068631	+0.000135813
-0.10	-0.067976	-0.000136940

At amplitudes 0.05/0.10 Angstrom, energy secants κ_E were -0.040317/-0.013661 eV/Angstrom squared, whereas gradient secants κ_g were -0.016823/+0.001364. They sample different properties outside the quadratic regime. A restoring gradient at larger displacement is compatible with energy remaining below the center; the sign difference alone neither proves energy-force inconsistency nor certifies an intervening stationary point.

A retrospective adjacent-interval work check guards against overinterpreting opposite-endpoint cancellation. Energy-minus-trapezoid-work residuals were +0.029245/-0.029491 meV/cell on the -0.05-to-0 and 0-to-0.05-Angstrom intervals; both outer interval magnitudes were below 0.000461 meV/cell. These quadrature discrepancies do not separate anharmonicity from numerical energy/gradient effects and are not total-error bounds. We did not shift the cached origin or refit coefficients to close them. This very shallow lowering is not evidence of a new phase or an improved whole-path barrier.

The four new statics cost 3.756877 completed-call proxy core-hours and 4.248889 allocation core-hours, with no new center SCF. The preceding 272 post-release derivatives cost 276.622222 allocation core-hours; earlier training and preparation remain additional. Training commitments are T34/81 and preserving39/81. At this four-probe training stage the sixteen original independent targets were uncomputed. The later T local test is reported in Appendix F.2; the revised preserving and physical +0.5% tests are now complete, with the latter reported in Section 3.4. Original admission refusal is preserved, not backfilled by these training results.

E.8. The original T local independent cohort

All eight originally frozen T targets were processed without a replacement point, start, model, derivative step or branch. Each released-gradient norm was between 0.005288 and 0.029500 eV/Angstrom. Both original allocations ended normally, after 837 and 1320 seconds at 32 MPI ranks; the complete read-only audit reparsed all eighteen native SCFs, their energy/force/stress records, input hashes and charged attempts. No new Hessian, NEB or center SCF was added.

Table A18 reports every registered arm. Errors are meV/HfO2, four formula units per twelve-atom cell. The first two aligned columns use each arm's eligible labels; n is its coverage out of eight, not its count of independent replicates. The common maximum uses the same four axial labels for all nineteen arms. Neither different coverages nor four equivalent Cmma registrations are statistical samples for a pooled method ranking. Probe steps are Angstrom.

Table A18. All frozen T test arms, coverage and both error conventions.

Model	n/8	Max. aligned	Aligned RMSE	Common max.	Max. unaligned
Bilinear	8	0.256803	0.186280	0.180268	0.256803
Tensor quadratic	8	0.089077	0.048248	0.053665	0.089077
Refitted total quadratic	8	0.105388	0.062391	0.105388	0.098433
Original H/g, 0.01	8	0.129769	0.060608	0.071940	0.076065
Original H/g, 0.02	8	0.131911	0.062199	0.073492	0.076339
T triplet, 0.01	6	33.075809	24.033465	33.075809	33.170243
T two optical doublets, 0.01	8	118.771513	63.020030	38.103950	118.866029
T all optical, 0.01	8	0.129769	0.060608	0.071940	0.076065
Cmma 103, 0.01	6	50.021808	33.963295	50.021808	50.114924
Cmma 106, 0.01	6	50.021575	33.962239	50.021575	50.114690
Cmma 114, 0.01	6	50.021575	33.962239	50.021575	50.114690
Cmma 127, 0.01	6	50.021808	33.963295	50.021808	50.114924
T triplet, 0.02	6	33.072958	24.031991	33.072958	33.167766
T two optical doublets, 0.02	8	118.767602	63.014235	38.109537	118.862499
T all optical, 0.02	8	0.131911	0.062199	0.073492	0.076339
Cmma 103, 0.02	6	50.012837	33.955505	50.012837	50.106338
Cmma 106, 0.02	6	50.012590	33.954441	50.012590	50.106090
Cmma 114, 0.02	6	50.012590	33.954441	50.012590	50.106090
Cmma 127, 0.02	6	50.012837	33.955505	50.012837	50.106338

The restricted references clamp directions outside their declared span at the same Q. Their errors are not a test of every native Cmma-coordinate model. The full T reference includes the entire declared internal space and is equivalent to the original full-H control, not an independent method replication. All raw/calibrated signed errors, origin biases, geometry abstentions and source identities remain in the test case. The eighteen test SCFs and their 19.173333 allocation core-hours are additional to all original preparation. The original preserving two-coordinate target remains refused; its separately revised eight-candidate test is reported below. The separately frozen physical +0.5% path test is complete and reported in Section 3.4. These local errors do not supply a whole-path barrier error or a resolved methodological gain.

E.9. The revised preserving local independent cohort

The complete eight-candidate native audit covers24SCFs and both original normal terminal allocations,1069/2075seconds at32CPUs. Every original start, per-point cap, native source and physical input was checked; no second start, retry or post-label threshold change was used. All candidates passed the registered dual-gradient conditions, but their transverse stability remains unchecked. The original two-coordinate refusal and the already seen T test remain separate evidence, not sixteen stable-sheet successes.

Table A19. All23frozen arms on revised preserving candidates (meV/HfO2).

Arm and source step/Angstrom	Supported/8	Raw max.	Raw RMSE	Aligned max.	Aligned RMSE
New full-space harmonic, 0.01	8	0.033195	0.026829	0.033195	0.026829
Retained nonlinear, 0.01	8	0.035625	0.024219	0.035625	0.024219
New full-space harmonic, 0.02	7	0.033928	0.027050	0.033928	0.027050
Retained nonlinear, 0.02	8	0.036390	0.024151	0.036390	0.024151
Bilinear	8	0.182055	0.144969	0.182055	0.144969
Tensor quadratic	8	0.029103	0.018169	0.029103	0.018169
Refitted quadratic	8	0.059872	0.044073	0.063879	0.035656
Original H/g, 0.01	8	1.611666	1.534980	0.352593	0.295880
Original H/g, 0.02	8	1.664167	1.585322	0.358942	0.300219
T triplet, 0.01	0	–	–	–	–
T two optical doublets, 0.01	8	147.755933	88.600341	135.387518	78.347208
T all optical, 0.01	8	1.611666	1.534980	0.352593	0.295880
Cmma 103, 0.01	2	19.448543	15.751533	7.214300	5.192788
Cmma 106, 0.01	2	19.264559	15.595606	7.030254	5.082366
Cmma 114, 0.01	2	19.264559	15.595606	7.030254	5.082366
Cmma 127, 0.01	2	19.448543	15.751533	7.214300	5.192788
T triplet, 0.02	0	–	–	–	–
T two optical doublets, 0.02	8	147.677582	88.557743	135.309194	78.304392
T all optical, 0.02	8	1.664167	1.585322	0.358942	0.300219
Cmma 103, 0.02	2	19.444749	15.747429	7.210592	5.190878
Cmma 106, 0.02	2	19.260955	15.591722	7.026735	5.080630
Cmma 114, 0.02	2	19.260955	15.591722	7.026735	5.080630
Cmma 127, 0.02	2	19.444749	15.747429	7.210592	5.190878

Errors are numerical discrepancies on each arm's available IDs; different support is not a common-domain comparison. Missing values are not zero errors. All four Cmma registrations and both source steps are retained.

The four new models share IDs1–7; target8 is the pre-label harmonic0.02 model-oracle budget refusal, not a failed DFT calculation. The original T triplet has no supported target and the Cmma restrictions each have only two; there is no all23-arm common target set. Original restricted-model numbers remain a new-target diagnostic, not retroactive stable-plane admission or failure of every possible native Cmma-coordinate model. Raw/aligned errors, all184predictions and all8native candidate energies are available through the offline paper-table command. The complete fixed scoring uses no new fit, bias, source-step selection or extrapolation across the failed training center. Neither the small errors nor the increased candidate coverage certify a stable surface, a TS or a physically resolved methodology gain. The completed physical +0.5% path comparison is reported separately in Section 3.4.

Appendix F. Reference coverage, branch trials and model-comparison details

F.1. Representation coverage and bounded local branch tests

A compact displacement projection did not imply a compact physical-gradient representation. At the zero-strain T escape peak the geometric T triplet covered 99.0957% of squared atomic displacement but only 14.3013% of squared internal atomic gradient. For the M peak the corresponding fractions were

57.9237% and 9.0282%. All four registered rank-four Cmma frames retained 93.5085–93.5087% of the M gradient and 54.0718–54.0765% of the T gradient. Low displacement coverage therefore did not disqualify Cmma as a strong energy-model control. The rank-three triplet and the rank-four T optical and Cmma spaces were not treated as dimension-matched models.

Figure A5 separates energy, displacement and gradient profiles in the same original-T mass chart. The twenty-seven records include cached endpoints, not twenty-seven independent replicates or new SCFs. Cell strains and physical-gradient norms are kept separate from atomic coverage fractions. The full eight-band peak coverage table is retained in Appendix D, Table A5.

The strain comparison retained the same reference metric, original T optical doublets, all four Cmma registrations and continuous ordered lifts (Figure A6). At the T candidate's peak the triplet displacement fraction remained 99.0957%/99.0045% at zero/+1%, while its gradient fraction changed from 14.3013% to 69.7245%. The corresponding Cmma fractions changed from 54.0718–54.0765% to 10.5750–10.5766%. The internal gradient norms were 0.010196606 and 0.038221369 eV/(sqrt(amu) Angstrom); the percentages alone are neither model errors nor a reference-transferability test.

The M pair provides a contrasting strong-reference diagnostic (Figure A7). At +1% its T-triplet displacement and gradient fractions were 54.9251% and 3.9008%; Cmma retained 91.5873–91.5896% of the gradient. Its internal norm was 0.035561531 eV/(sqrt(amu) Angstrom). Thus the poor geometric-T gradient coverage cannot establish superiority over the stronger fixed reference.

The two ordered flip candidates probed different regions of the same landscape (Figure A8), not the globally easiest homogeneous switching mechanism. At zero strain their raw peaks were 96.828183 and 338.968965 meV/HfO2. The preserving peak is not a certified index-one saddle. At that peak the triplet covered 82.6301% of displacement but 12.9583% of gradient; the complete optical rank-four control was below 0.0001% for both, whereas Cmma retained 52.7203% of the gradient. At the reversing peak the triplet gradient fraction was 0.0320%, compared with 65.1046% for Cmma. Parent-equivalent minus endpoints remain different at ordered indices; their phase equivalence does not establish whole-path equivalence or electronic winding.

Eight of eight registered G2 training bands are ordinary-converged. These descriptive projections neither select a reference winner nor certify conditional energies, transverse stability or independent prediction. Appendix D retains every reference rank, all four tied Cmma frames and their nonstatistical registration ranges. The original source modes and forces are not modified by ASR; raw translation contributions remain visible.

For the preserving local cohort, most of the energy lowering was already obtained by atomic rearrangement at each frozen-Q seed cell. Across the five registered coordinates, atomic-only release lowered the energy by 44.06–48.35 meV/cell. The joint candidates were a further 0.80–1.14 meV/cell lower. The origin-aligned atomic-only model discrepancies were 4.349824 and 4.399770 meV/cell for the two original probe steps, compared with 4.671404 and 4.720758 for joint release on the same four axial points (Appendix D.8). Thus the local model mismatch was already largely present before the additional joint-release step. Omission of open-cell release alone cannot explain it. These observations cover the specified local candidates, not the whole switching path or a unique microscopic energy partition. All ten atomic-only candidates passed their own gradient target but retained joint released-gradient norms above that target; neither joint stationarity nor transverse stability followed from this test.

The post-release curvature test exposed a specific omitted direction at the preserving origin. Its transverse block retained one raw negative eigenvalue at both steps, approximately -0.0221/-0.0241 eV/Angstrom squared, despite meeting the released-gradient target. The directions had squared overlap 0.999998 and were almost entirely oxygen displacements in the registered length metric. Freezing the cell did not remove the raw negative direction. Four same-contract statics along it lay 0.0126–0.0172 meV/HfO2 below the cached origin (Appendix E.7). This was a shallow local lowering, not a corrected whole-band barrier or an explanation of the larger release-shape discrepancy. The original two-coordinate conditional admission remained refused; formal retention of a third coordinate alone did not certify a nonlinear branch.

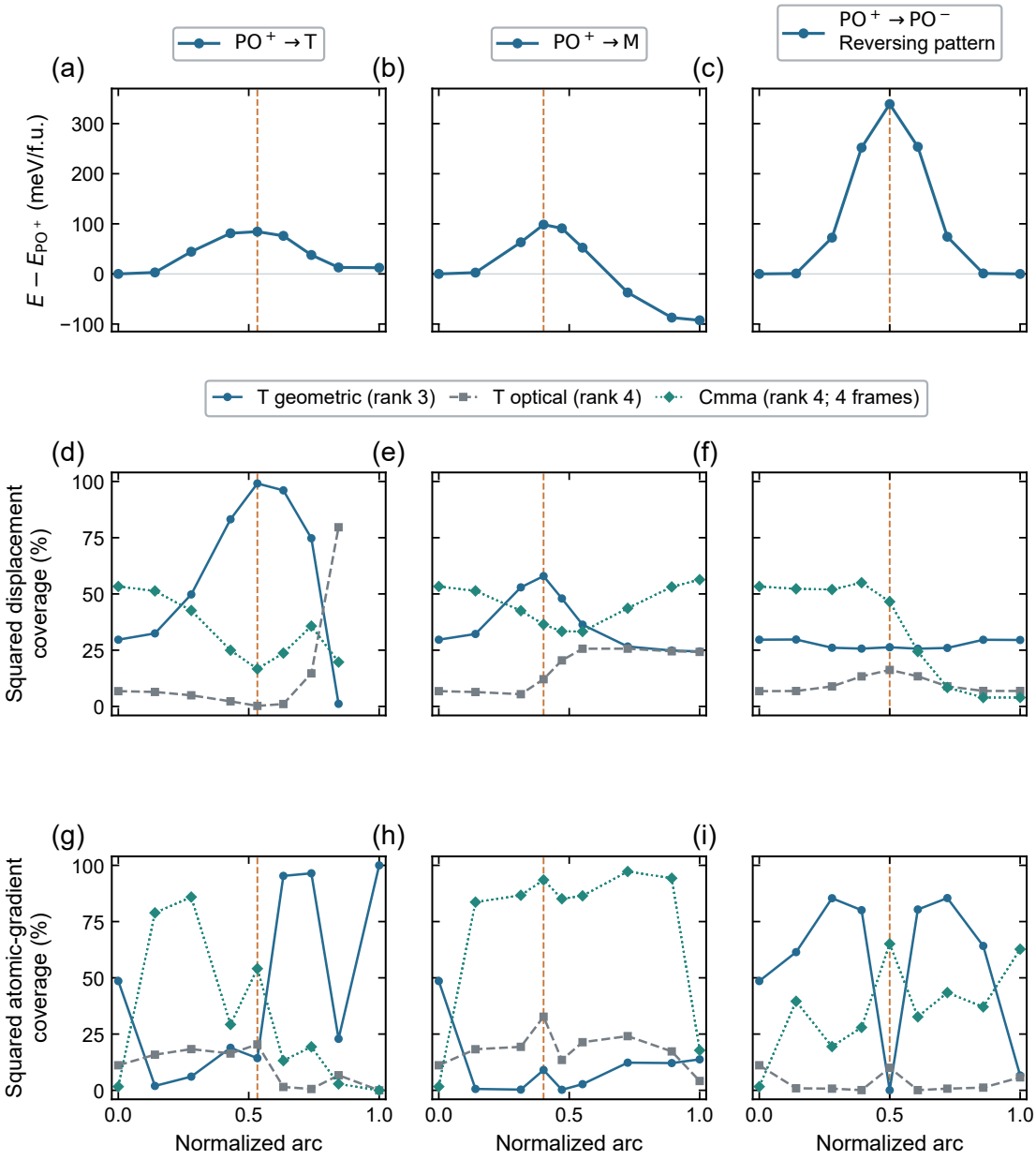


Figure A5. Three ordinary-converged zero-strain training candidates: PO-to-T (left), PO-to-M (middle) and reversing-pattern switching (right). (a–c) Nine P=0 energy samples per chain on one energy scale, relative to the same cached PO+ state. (d–f) Squared translation-free displacement coverage in the common T chart. (g–i) Squared dual atomic-gradient coverage, with different denominators from (d–f). Blue, grey and teal denote the T geometric triplet (rank3), two complete T optical doublets (rank4) and all four Cmma registrations (each rank4), respectively. Orange dashed guides mark sampled peaks, not certified TSs; connectors do not imply continuous MEP interpolation. The 27 records are descriptive samples, not independent replicates; registration spread is not a statistical uncertainty interval.

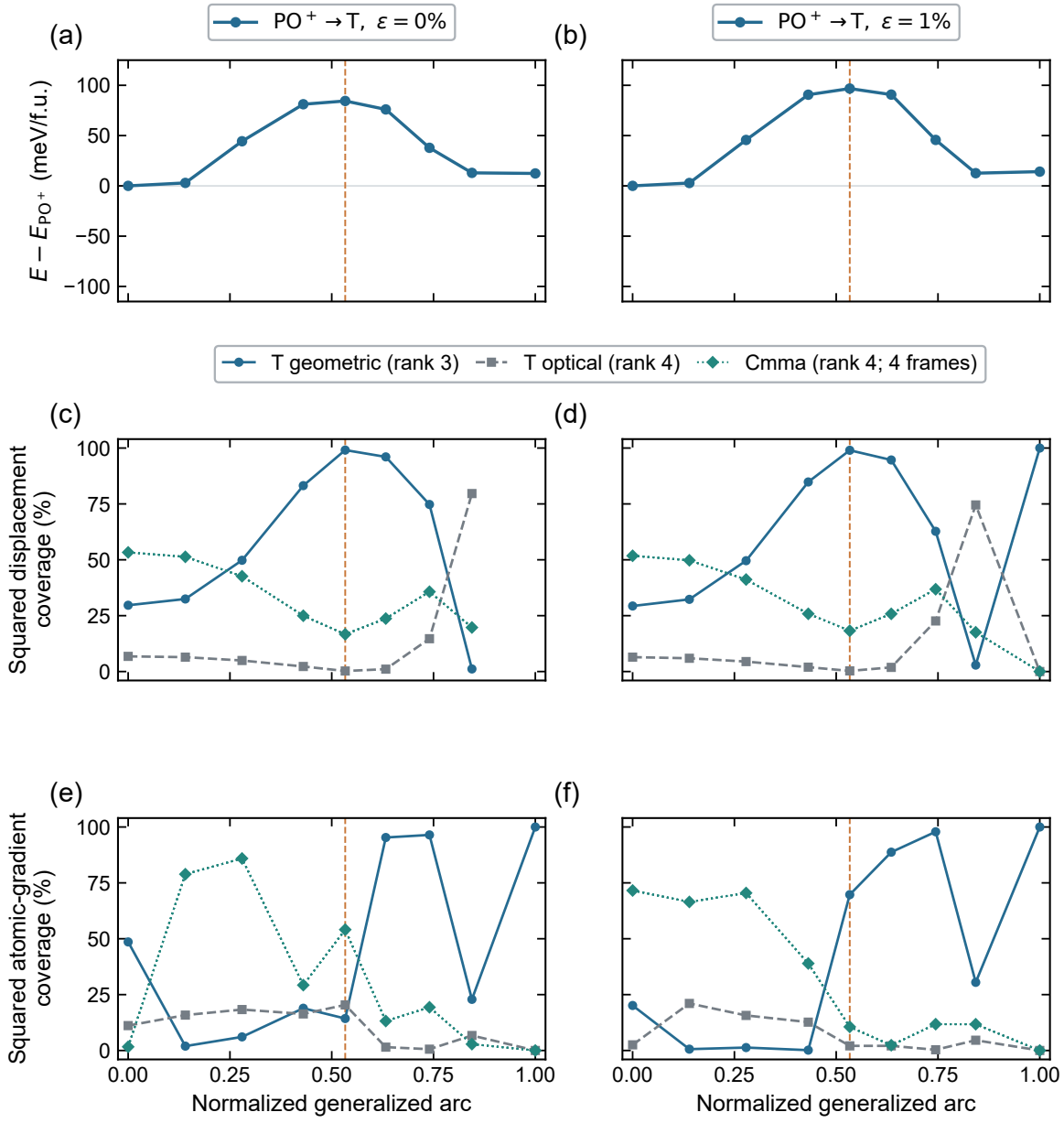


Figure A6. Zero (left) and +1% (right) substrate-strain training bands. (a,b) Energies relative to each matched PO+ endpoint. (c,d) Squared displacement coverage in the same unstrained-T chart. (e,f) Squared dual atomic-gradient coverage. All four Cmma registrations are retained, not an uncertainty band. Guides mark highest sampled images; connectors are not a fitted MEP or pointwise inter-strain correspondence.

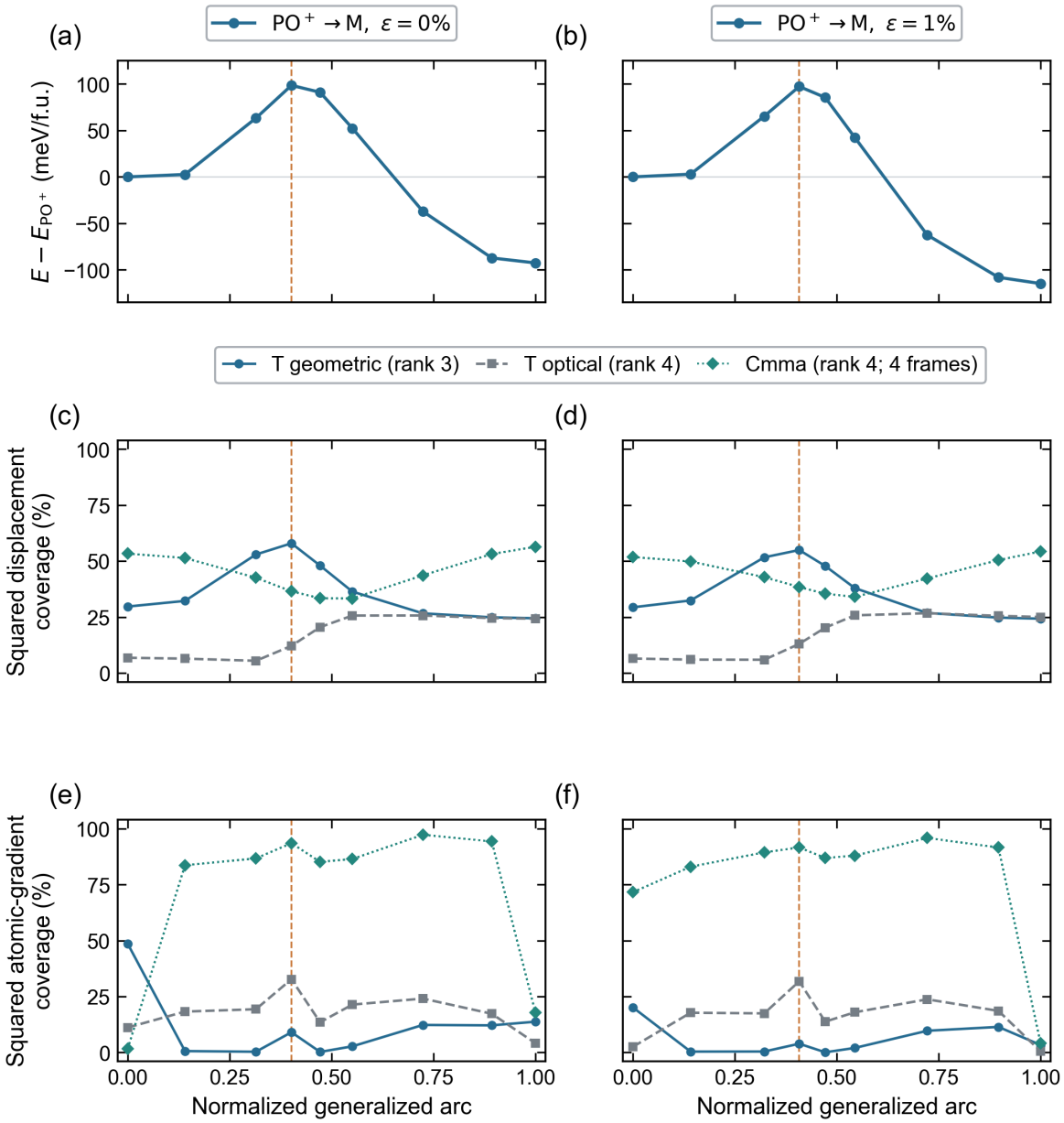


Figure A7. M escape training bands at zero (left) and +1% (right) substrate strain. (a,b) Native energies relative to each matched PO+ well. (c,d) Squared translation-free atomic-displacement coverage in the same original T mass chart. (e,f) Squared atomic-gradient coverage in the corresponding dual chart. Blue circles denote the rank-three geometric T triplet, grey squares the two complete rank-four T optical doublets, and green dotted curves all four preregistered rank-four Cmma frames. Vertical dashed lines mark discrete highest images, not certified TS locations. Each band has its own normalized generalized arc; equal abscissae do not match configurations. The eighteen rows include shared cached endpoints and are not independent replicates. Cmma-frame spread is not a statistical interval. No smoothing, fitting, new mode selection or physical input change is applied. Scalar source data and editable vector exports accompany the figure contract.

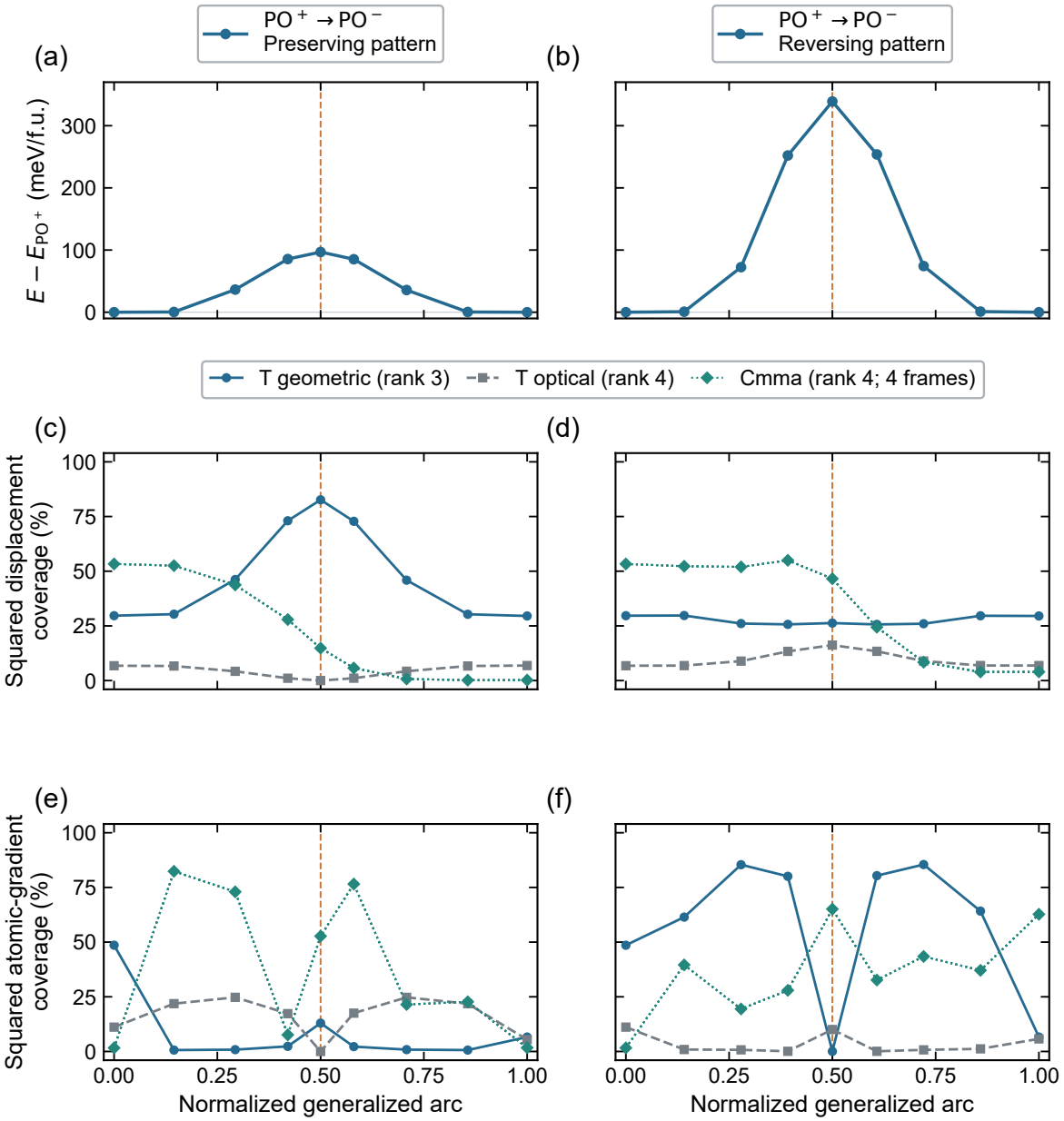


Figure A8. Two ordinary-converged zero-strain ordered flip candidates, preserving (left) and reversing (right). (a,b) Native sampled energies, with the same PO+ zero and energy scale. (c,d) Squared displacement coverage in the common T chart. (e,f) Squared dual atomic-gradient coverage in that chart. Both T reference ranks and all four Cmma registrations are retained. Orange guides mark highest sampled images, not certified TSs. Each column has its own normalized generalized arc; equal abscissae are not matched configurations. Eighteen frame records share the initial native cache and are not independent replicates. Straight connectors are not a smooth-MEP reconstruction, and frame spread is not an uncertainty band.

Allowing that direction to relax explicitly did not complete the central branch within the registered search domain. Seven of the ten two-sided training starts met both gradient tests in 81 native SCFs altogether. Both Q=0 starts approached the original 0.5-Angstrom total-displacement bound with released-gradient norms near 0.296 eV/Angstrom. The third capped start was not radius-limited: its norm was 0.001166 eV/Angstrom, but its directional gradient, 1.30035×10^{-4} eV/Angstrom, exceeded the preset requirement. Five energy-rejected trials consumed its remaining allowance. These distinct failures were retained rather than repaired by changing the geometry or electronic settings. At the three Q positions where both starts passed, the resulting structures differed by about 0.0028 Angstrom and less than 0.000607 meV/HfO2. They did not establish two distinct stable branches. The complete trial and native replay retain all ten outcomes; their 85.6 allocation core-hours are additional to the earlier curvature and sampling costs. The original two-coordinate refusal is unchanged, and no continuous stable three-coordinate surface is inferred from these candidates.

F.2. Complete prediction controls and local independent errors

An independent prediction advantage has not yet been established. A complete T optical reference plus the permitted cell directions represents the same full quadratic as the coupled-coordinate model: changing that complete basis did not change its predictions (Appendix E.5). A useful comparison must therefore distinguish nonlinear energy modeling and branch validity from a relabeling of the same response. The energy-only controls below address that comparison on already seen data; they do not supply the independent test.

A standard energy interpolant described the seen T fine-node energies more closely than the unchanged original-center quadratics. All five arms supported the same sixteen fine targets (Table A20). The tensor-quadratic maximum shape discrepancy was 2.003468 meV/cell, compared with 12.901228 and 12.975249 meV/cell for the two original H/g models. This restricted cohort differs from the twenty-four nonzero targets in Table A16; the smaller quadratic maxima do not represent a model improvement.

Table A20. Post-hoc T controls on sixteen common fine targets (meV/cell).

Model	Max. shape discrepancy	Shape RMSE
Bilinear	6.900894	4.219064
Tensor quadratic	2.003468	1.043743
Refitted total quadratic	7.667090	4.006016
Original H/g, 0.01 Angstrom	12.901228	5.177484
Original H/g, 0.02 Angstrom	12.975249	5.219402

Origin calibration matters even for this retrospective comparison. The refitted total quadratic had an origin bias of +1.546418 meV/cell; its maximum unaligned discrepancy was 6.120672 meV/cell. We retained that quantity as well as the origin-aligned values rather than choosing the more favorable metric. The interpolants used coarse energy labels, whereas the original quadratics were not fitted to those energies. Their different information and preparation do not support an equal-budget advantage.

The preserving sheet did not support the same two-dimensional test. Only five of its nine coarse labels were available, leaving the six-term quadratic design rank five. The fit was therefore refused, without a pseudoinverse or invented corner energies. No fine target was common to all five arms; the remaining arms shared only four axial targets. That axis-only diagnostic did not validate a two-dimensional surface.

This was not an independent forecast or a saving of DFT calls. Every label was already seen, and all eight nonzero coarse starts at each center inherited fine-node parents. Historical coarse/fine-node SCF counts were 47/71 for T and 58/114 for preserving; unsuccessful nodes remained charged. The original curvature, Gamma, canary and G2 preparation were additional. The complete coarsening record retains raw and aligned errors, missing nodes and branch-start provenance. Its implication is a necessary comparator, not

a new interpolation method: nonlinear reduction must be tested against a competent energy-only model, not only against an unrefitted local quadratic.

The prospective T local test supplied eight independent gradient-candidate energies, using every original nonzero point of the central $\{-0.05, 0, 0.05\}$-Angstrom square. The coordinates, single Q0-parent starts, five standard arms and fourteen reference/step records were frozen before these labels. Both T-origin raw transverse blocks had passed the original necessary sign screen; no test point or electronic input was changed. All eight candidates reached the unchanged 0.03 eV/Angstrom released-gradient target in eighteen native SCFs. Table A21 reports the predeclared origin-aligned errors and retains the unaligned maximum rather than choosing a favorable zero.

Table A21. Prospective T local errors on eight common targets (meV/HfO2).

Model	Max. aligned error	Aligned RMSE	Max. unaligned error
Bilinear	0.256803	0.186280	0.256803
Tensor quadratic	0.089077	0.048248	0.089077
Refitted total quadratic	0.105388	0.062391	0.098433
Original H/g, 0.01 Angstrom	0.129769	0.060608	0.076065
Original H/g, 0.02 Angstrom	0.131911	0.062199	0.076339

All five maximum aligned errors were below the preregistered 1 meV/HfO2 engineering reporting target, which is not a physical-error bound. The tensor interpolant had the smallest aligned maximum; the original quadratics had smaller unaligned maxima. There was therefore no metric-independent winner. The complete T-optical reference reproduced the corresponding full-H errors, as expected from the same complete quadratic. Restricted triplet, two-doublet and Cmma controls were also retained, including their geometry abstentions (Appendix E.8, Table A18). All nineteen arms shared only four axial test points; that common subset alone was not a two-dimensional validation.

These eighteen new SCFs cost 17.833999 completed-call proxy core-hours and 19.173333 allocation core-hours; original curvature, training and branch preparation remain additional. The test checked one prescribed single-start gradient-candidate branch, not global conditional minima or its full transverse stability. It did not establish an advantage for a new reduction method. The complete native local test and fixed scoring retain every target, both error conventions, all nineteen arms and actual cost.

The retained-direction nonlinear fit improved training energies without establishing a transferable accuracy gain. Its maximum training errors were 0.1769/0.1686 meV/HfO2 at the two source steps, versus 3.6167/3.5844 for the same-information harmonic family. Excluding the central Q family reversed that comparison: nonlinear maxima were 4.0836/4.4101, versus 3.6167/3.5844. This exclusion also changed support, from training amplitudes $|\zeta| \leq 0.1000$ Angstrom to test amplitudes as large as 0.3688 Angstrom. It was therefore a shifted-support training diagnostic, not the independent small-Q interpolation test. All fitted and group-excluded designs were identified at the fixed cutoff; maximum nonlinear training full-gradient errors remained about 0.213 eV/Angstrom. Energy-fit improvement alone did not validate gradients, a stable conditional branch or a new method advantage.

The preserving origin still fails the original necessary sign condition. After the finite training trial, a separately registered test froze all four family/step predictions for the original eight remaining Q positions, using one fixed training-parent start per position and the same dual-gradient stopping rule. There were 31 numeric forecasts and one explicit twelve-call model-oracle refusal; those oracle calls were not DFT costs. All nineteen older controls, their reference definitions and the original two-coordinate refusal remained in the panel. The revised target is a single-start gradient-candidate energy, not retroactive stable-sheet validation. These pre-label forecasts were tested without refitting. All eight candidates passed both gradient tests in 24 native SCFs, with 2/2/2/2/6/4/3/3 calls in original target order. Released norms were 0.00164–0.01949 eV/Angstrom and the largest absolute directional gradient was 9.918×10^{-5} eV/Angstrom. Full native energy/forces/stress, unchanged physical inputs and genuine32MPI were reparsed

for every charged call. The complete audit and fixed scoring retain all 23 arms and 184 model/point records. This completes the revised candidate-label test, not stable-sheet validation. The original sixteen-point inventory now contains eight original T labels and eight revised preserving labels; the original preserving sign-screen refusal is not reversed.

Table A22. Revised preserving errors on the seven targets supported by all four new family/step arms, in meV/HfO2. Coverage refers to the original denominator eight. Raw and origin-aligned values coincide for these four fixed-origin models; no independent-label calibration was applied.

Family	Probe step/Angstrom	Coverage	Common-set max. error	Common-set RMSE
Full-space harmonic	0.01	8/8	0.033195	0.027153
Retained nonlinear residual	0.01	8/8	0.035625	0.024916
Full-space harmonic	0.02	7/8	0.033928	0.027050
Retained nonlinear residual	0.02	8/8	0.036390	0.024852

The nonlinear family had slightly smaller RMSE but larger maximum error at both steps; neither metric supplied a physically resolved advantage. The one harmonic model-oracle budget refusal remained missing even though its corresponding DFT candidate converged. All nineteen old controls were also scored numerically against the revised target (Appendix E.9), without treating that agreement as validation of their old stable-plane assumptions. Their tensor interpolant had maximum raw discrepancy 0.029103 meV/HfO2, comparable to the new families, but used a different previously registered training representation. Across all 23 arms the common supported set was empty; restricted references were not compared as if they predicted a common full surface. The test cost 26.469276 completed-call proxy core-hours and 27.946667 allocation core-hours. All 81 training SCFs, both curvature groups and earlier preparations remain additional. These small local errors do not establish a more accurate force field, stable sheet, stationary TS or unseen-strain path prediction. The separately frozen +0.5% mechanical path test is now complete and reported in Section 3.4; local-coordinate accuracy remains a different observable from its physical path response. The original local forecasts and strong-reference forecasts remain unchanged, including the refused preserving cohort.

Appendix G. Response theory and extended computational details

G.1. Mechanical ensembles and branch-aware reduction

All images and endpoints of one path share the same permitted cell space. The epitaxial calculation fixes two T-referenced in-plane vectors and releases the third vector, including its two tilt components. Atomic and cell variables are relaxed in that same space. A clamped reaction stress is not an open-coordinate convergence failure. Zero applied stress and zero nominal substrate strain are distinct ensembles, and are not adjacent points of a single strain derivative.

Adding a constraint orders ideal minimax barriers only when the endpoints and thermodynamic potential are held common. For allowed continuous paths $\mathcal{G}_U(I,F)$ and $\mathcal{G}_C(I,F) \subseteq \mathcal{G}_U(I,F)$,

$$B_U^*(I,F) = \inf_{\gamma \in \mathcal{G}_U(I,F)} \max_{s\in[0,1]} E(\gamma(s)) - E(I) \le B_C^*(I,F).$$

The star denotes the ideal minimax value, not a finite-image NEB estimate. Neither a local optimizer nor discrete sampling establishes that ordering for two computed candidate bands. More importantly, newly relaxed endpoints under a different substrate do not satisfy the common-endpoint premise. Our free-cell and T-substrate-clamped results therefore are not interpreted as a nested-path inequality, and the matched 0/+1% clamped comparison still requires each condition's own initial-well subtraction. A higher absolute bottleneck energy can accompany a lower activation barrier if the well rises more. Appendix B gives an elementary counterexample, not hafnia evidence or a new theoretical contribution. This is why the

endpoint-response controls and the separate absolute escape-barrier test cannot be replaced by a generic statement that clamping raises barriers.

In the registered generalized-coordinate metric, a prescribed active-space projector separates the allowed physical gradient from support reactions. Ordinary NEB residuals, allowed physical gradients and excluded reactions are reported separately. Only enforced mechanical constraints enter this projection; an unmeasured but physically movable mode is not relabelled as a support reaction merely because a reduced model omits it.

A retained coordinate q can specify one mode or a declared combination of modes. A frozen slice keeps the remaining coordinates r fixed. A conditional surface instead follows a particular local minimum in the permitted r space at each q; the lower envelope over branches is a third object. With q and r measured from the same expansion centre and stable H_{rr}, local quadratic release gives

$$r^*(q) = -H_{rr}^{-1}(g_r + H_{rq}q), \qquad K_{\text{eff}} = H_{qq} - H_{qr}H_{rr}^{-1}H_{rq}.$$

The offset from a nonzero g_r is retained. The Schur complement is standard mathematics, not a new theorem. An unresolved or unstable eliminated block invalidates this release; a physical unstable direction cannot be removed by a pseudoinverse. Unmeasured directions remain fixed, not implicitly stable. At a smooth constrained stationary branch, the corresponding barrier strain derivative is the difference of bottleneck and initial-state work derivatives, evaluated with their actual cell Jacobians and volumes. An ordinary NEB sampled peak alone does not justify that stationary-branch formula.

G.2. Controlled stationary response and paired energy gaps

Stable orthogonal release and continuation of a stationary bottleneck are different operations. After the permitted stable r block has been eliminated, write the local energy in retained internal coordinates q and a prescribed external coordinate t as

$$\begin{aligned}\mathcal{E}(q,t) = \widetilde{E}_0 + \widetilde{g}_q^T q + \widetilde{g}_t t \\ + \tfrac{1}{2}q^T A q + t b^T q + \tfrac{1}{2}dt^2.\end{aligned}$$

All quantities come from one centre, potential and allowed-space chart. The t coordinate is controlled by the substrate, not released or counted as an internal unstable direction. If A is resolved and nonsingular, its specified stationary branch is

$$q^*(t) = -A^{-1}(\widetilde{g}_q + bt), \qquad C = d - b^T A^{-1} b.$$

In the length-valued response chart, both q and t have Angstrom units and A, b and d have units of eV Å^{-2} at the cell-energy level. Thus $dq^*/d\epsilon = -LA^{-1}b$ has Angstrom per unit strain, not the energy-density units of an elastic stiffness. These coordinates are not the mass-weighted amplitudes used for path projections. An acoustic continuum relation or a fitted optical-mode slope from a different reference/boundary does not independently validate this internal mixed response. Same-contract probes and prospective material tests must do so; the physical-strain comparison is completed in Section 3.4, without a validated transfer of these local stationary models.

For an internal minimum, A is positive definite and C cannot exceed d. For an index-one saddle, A is indefinite and this curvature correction is not sign definite. A bottleneck's controlled curvature can therefore harden even though elimination of its stable directions softens a retained block. This follows from standard quadratic stationary response, not a new theorem; softening of one selected mode alone does not determine a barrier change. An unresolved inverse is rejected rather than repaired by a pseudoinverse. The index is established only in the represented internal space; physically movable but unrepresented gradients remain distinct from clamped reactions.

Each centre a uses its own registered scale $t_a = L_a(\epsilon - \epsilon_0)$. Let e_a(t) denote its stationary model energy change relative to its raw centre energy, including all nonstationary anchor corrections. With the independently recorded raw centre gap $G_{\text{ref}} = E_{S,\text{ref}} - E_{I,\text{ref}}$, define

$$D(\epsilon_0) = \frac{G_{\text{ref}} + e_S(0) - e_I(0)}{n_{\text{fu}}}.$$

The small increments $d_a = e'_a(0)t_a + C_a t_a^2/2$ avoid subtracting large total energies to obtain response. The paired change and its derivatives are

$$\delta D = \frac{d_S - d_I}{n_{\text{fu}}}, \qquad D(\epsilon) = D(\epsilon_0) + \delta D,$$

$$\frac{dD}{d\epsilon} = \frac{L_S e'_S(t_S) - L_I e'_I(t_I)}{n_{\text{fu}}}, \qquad \frac{d^2D}{d\epsilon^2} = \frac{L_S^2 C_S - L_I^2 C_I}{n_{\text{fu}}}.$$

Consequently neither equal numerical t values in different charts nor a bottleneck response with the initial-well term omitted represents this mechanical intervention. The raw gap, stationary-anchor correction and parameter response are retained separately. A common energy-zero shift does not affect the response; negative model gaps are not clipped into apparent activation barriers.

D equals a physical activation barrier only when I and S are the appropriate actual stationary initial state and connected bottleneck on the same branch. A restricted quadratic solution cannot establish those premises, identify the highest bottleneck of a complete path or exclude lower competing routes. Adding δD to an audited NEB barrier is a separate calibration choice: its anchor mismatch must be disclosed and checked, not silently discarded. Caller-declared parameter bounds and displacement radii reject obvious out-of-domain use but do not prove probe-hull coverage or anharmonic accuracy. The stationary and paired-response implementations and their analytic tests are thus methodological preparation, not new hafnia barrier data or a successful independent material prediction.

A full initial-well quadratic need not be constructed for every comparison. When audited target endpoint energies are declared visible inputs, the bottleneck model can instead be paired with that measured initial-well shift. Every control receives the same endpoint information and its preparation cost. This is endpoint-assisted bottleneck prediction, not a zero-DFT or fully predicted well response. It does not add an unregistered initial-well Hessian to the finite two-bottleneck probe budget. The exact input choice, basis, domain and calibration must be frozen before complete target path labels are read; the present module tests alone do not freeze that batch.

The local ablations now share one dataset of measured physical Hessian actions $D = HM$, where the columns of M are orthonormal in the declared joint coordinate metric. A control is unavailable unless its retained, released and prescribed-control directions lie in this measured span; an unmeasured complement is unknown, not a stable zero block. Frozen, atomic-only and joint-release controls retain the same affine gradient. Atomic-only release is unavailable if the retained coordinates already contain cell motion. The local stability-aware arm promotes the complete released eigensubspace at or below the registered resolution floor, then requires a resolved combined internal block of the specified index. This is training-only increment-dimension continuation, not an anharmonic branch search. When joint release is already valid, promotion changes no physical stationary response and cannot establish an accuracy advantage. Measured coverage, strong reference registration and material validation remain separate gates.

For the same complete quadratic and admissible internal space, valid joint stationarity is also invariant under changes of the retained subspace and its dimension. Direct full-space solves verify this null across five partitions, including a correctly refused unstable-release case restored by promotion. This prevents a coordinate-partition gain from being claimed as physical prediction improvement. Independent nonharmonic, branch or equal-accuracy cost evidence is needed to establish an actual increment; a surface at fixed retained coordinates is a different object. The partition check is not a hafnia material forecast.

G.3. Signed path work as an energy-consistency diagnostic

Displacement and instantaneous gradient coverage do not measure the energy change accumulated along a path. We therefore retain a separate, signed physical-work check on the sampled coordinate polygon. For successive coordinates x_i and their conjugate physical energy gradients g_i, define

$$W_i = \frac{g_i + g_{i+1}}{2} \cdot (x_{i+1} - x_i), \qquad R_i = E_{i+1} - E_i - W_i.$$

Within each declared coordinate block, orthogonal retained, omitted and translation projectors partition this trapezoidal work exactly. Atomic mass coordinates and the original permitted open-cell coordinates retain their own dual gradients and units; no basis mixes these blocks. The cell contribution completes the physical contraction. Rotating a basis inside one span changes no result, but choosing another span changes its work. Neither spring nor projected NEB forces replace the physical gradient.

These contributions are path-dependent signed work, not state-function energies assigned to phonon modes. Negative work is retained. The cumulative residual $\sum_i R_i$ diagnoses sampled energy-gradient consistency, but does not separate unsampled nonlinearity from force/stress error or bound the activation-barrier error. In particular, cancellation can make final endpoint closure small while the bottleneck-region discrepancy is large. The implementation and finite-data checks introduce no new calculator evaluations or reduction-model accuracy claim.

G.4. Work between the two seen strain conditions

To distinguish imposed control from internal configuration changes, the cross-strain diagnostic uses a single ordered zero-strain PO+ chart $y = (u, a, t)$. Atomic displacements u, the three open-row coordinates a and $t = L_\epsilon \epsilon$ are length-valued; $L_\epsilon = 5.12968067458423$ Angstrom. They are not the mass-weighted phonon amplitudes in the projection figures. The first two cell rows follow the prescribed substrate scaling; the original third row remains open. Physical gradients retain both internal forces and substrate reaction stress.

For equal-index native frames at the two training conditions, define

$$W_i^{\text{cross}} = \tfrac{1}{2}[g_i(0) + g_i(0.01)]^T[y_i(0.01) - y_i(0)], \qquad w_i = W_i^{\text{cross}} - W_I^{\text{cross}}.$$

The residual is $\Delta[E_i - E_I] - w_i$. Orthogonal atomic, explicit rigid- translation, open-row and prescribed-control blocks partition w exactly. The row-cell/fractional-coordinate connection, ordered lift, energy unit and initial-pair subtraction are fixed for every channel; no minimum-image convention (MIC) relabelling or intermediate DFT geometry is introduced. Equal image index is an explicit discrete pairing, not a canonical correspondence of continuous MEPs. The observed peak indices must coincide before this energy contrast is identified with the difference of sampled maxima.

These endpoint trapezoids are connection-dependent work, not unique mode energies, a frozen-slice measurement or a stationary barrier derivative. Their closure residual is not a total DFT or activation-barrier uncertainty bound. The two-configuration decomposition tests what a stress-only account omits without assigning a causal microscopic coupling.

G.5. Atomic/open-cell restrictions and finite-coordinate model checks

The same measured internal matrix is restricted to the 33-dimensional translation-free atomic space and the three-dimensional open-cell space to distinguish their raw instabilities. The prescribed substrate is not part of either internal instability count. Let P_a project onto the open-cell space in the fixed length metric and P_- onto the entire negative eigenspace of the symmetrized internal matrix. We report $W_a = \mathrm{Tr}(P_a P_-)$

rather than assigning individual vectors across nearly degenerate eigenvalues. This trace is invariant under orthogonal re-expression of these spaces, including negative-space basis rotations, but not arbitrary changes of the cell length scale. It is neither an energy fraction nor a phonon amplitude. A raw-positive cell block permits a formal cell-stationary Schur reduction; its nonzero cell gradient and constant energy offset are retained, as in Appendix G.1. Raw signs do not supply a measured stability-resolution floor.

We separately test whether an unchanged original-center quadratic describes finite-coordinate DFT candidates. Both finite-difference models use the same retained plane selected from the first-step matrix; no coefficients are fitted to the candidate energies. To isolate shape disagreement from the origin offset, we evaluate

$$d(q) = [E_{\text{model}}(q) - E_{\text{model}}(0)] - [E_{\text{candidate}}(q) - E_{\text{candidate}}(0)].$$

Subtracting the already seen candidate origin makes this a retrospective shape diagnostic, not a calibration-free or independent prediction. The common six-coordinate cohort is $(0, 0)$, $(-0.1, 0)$, $(0, -0.1)$, $(0, 0.1)$, $(0.1, 0)$ and $(-0.2, 0)$ Angstrom. The audited source snapshot contains nine T and six preserving candidates, all retained in the analysis. Equal numeric coordinates at the two centers do not identify the same physical structure. Finite released residuals also preclude treating retained partial gradients as exact conditional-envelope derivatives.

For the separate joint length chart, a fixed orthonormal partition $x = x_0 + V_q q + V_r r$ permits the work check of Appendix G.3 on neighboring conditional candidates. With $g_q = V_q^T g$ and $g_r = V_r^T g$, the endpoint trapezoids obey

$$\Delta E - W_q = W_r + \mathcal{R}, \qquad W_\alpha = \tfrac{1}{2}(g_{\alpha,i} + g_{\alpha,j})^T \Delta\alpha, \quad \alpha \in \{q, r\}.$$

Here $\mathcal{R} = \Delta E - W_q - W_r$ is the full endpoint quadrature discrepancy, not an energy-error bound. The Cauchy inequality bounds only the released endpoint term, $|W_r| \le (\|g_{r,i}\| + \|g_{r,j}\|)\|\Delta r\|/2$; it does not bound unsampled energy variation. We retain this term at finite convergence rather than setting it to zero by an exact-envelope assumption. Closed grid loops provide a complementary diagnostic: their raw energy changes telescope to zero, whereas finite endpoint-gradient quadrature need not. Neither a nonzero circulation nor a large neighboring displacement alone establishes nonconservative DFT forces or a branch discontinuity.

The matched frozen-R comparison separates two parts of an origin-aligned shape discrepancy without refitting either quadratic. For each original center, R=0 at four axial coordinates $q = (-0.1, 0)$, $(0, -0.1)$, $(0, 0.1)$ and $(0.1, 0)$ Angstrom supplies the frozen DFT counterpart. Both original finite-difference models use the same retained plane and affine gradient. Writing d_F and d_R for the frozen and released shape discrepancies, respectively, define

$$\delta_{\text{rel}}(q) = [E_R(q) - E_F(q)] - [E_R(0) - E_F(0)], \qquad d_R(q) = d_F(q) + \delta_{\text{rel}}^{\text{model}}(q) - \delta_{\text{rel}}^{\text{DFT}}(q).$$

Every term remains signed and is evaluated at the same q. The original raw and released origins are exactly reused. Because Q includes open-cell motion, R=0 is not a fully fixed-cell experiment. Positive released-minus- frozen candidate differences would be retained rather than clipped; a finite-gradient candidate is not assumed to be a global conditional minimum.

We additionally released the 31 atomic directions orthogonal to the same retained observables, while fixing each frozen-Q seed's cell. This atomic-only treatment, A, was compared with the frozen treatment F and the joint atomic- and-open-cell treatment J at the origin and four axial coordinates per center. Both original quadratic models were fixed before the atomic labels were read. The signed differences $E_A - E_F$ and $E_J - E_A$ were retained pointwise, along with each treatment's own origin. The second difference includes atomic readjustment as well as cell motion; it is not a unique strain-energy term. Because Q contains cell motion, A does not impose one shared cell across Q.

Reference comparisons at fixed coordinates must preserve the observables, not merely reuse their numerical labels. Let x denote the 36 internal coordinates, let $q = C^T x$ specify the same two retained observables in every arm, and let the orthonormal columns of U span the permitted displacement space. For $x = Uz$, the affine quadratic $g^T x + x^T H x/2$ is stationary under these constraints when

$$\begin{pmatrix} U^T H U & U^T C \\ C^T U & 0 \end{pmatrix} \begin{pmatrix} z \\ \lambda \end{pmatrix} = \begin{pmatrix} -U^T g \\ q \end{pmatrix}.$$

The constrained minimum requires positive curvature on the null space of $C^T U$, not positive curvature on all of U. We retain the nonzero affine gradient and reject a nonpositive free block rather than pseudoinverting it. If U spans the complete internal space, a different orthonormal reference is a re-expression of the same constrained problem. Its energy and physical coordinates must agree with an equivalent Schur solve. This standard constrained-quadratic identity is a comparator check, not a new theorem or a source of prediction gain.

The zero sign threshold used for this raw algebraic comparison is not a measured physical stability-resolution floor.

For restricted-reference ablations, U instead contains the transported atomic reference span and all three open-cell directions. Directions outside U are clamped; this is not the full outside-reference release of Appendix D.5. Each resulting geometry is checked against the same original 0.5-Angstrom radius and distance guard. We report the actual eligible coordinate sets and their intersection before comparing errors. Values outside the domain remain algebraic diagnostics, not failed physical predictions. These shared-observable ablations do not test a model defined with different native parent-phase coordinates and a different target.

To compare the local quadratic with elementary nonlinear energy models, we also performed a post-hoc coarsening of each original 5x5 sheet. The nine nodes on the 3x3 subgrid with coordinates -0.2, 0 and 0.2 Angstrom were designated coarse; the other sixteen nodes were designated fine. Energy-only controls were piecewise bilinear interpolation, tensor-quadratic interpolation (degree at most two in each coordinate), and a six-parameter least-squares polynomial of total degree at most two. The tensor polynomial can include mixed terms of total degree four; it is not another harmonic model. Both unchanged H/g quadratics were retained. Comparisons used only fine-node cohorts supported by all included arms, with the origin-aligned discrepancy defined above. This split was chosen after seeing the training labels. Moreover, coarse-node relaxation starts had inherited fine-node geometries; withholding fine energies from a fit did not blind the preceding workflow.

We checked the omitted preserving-center direction with four additional training statics, keeping the electronic contract unchanged. Let $x(\zeta) = x_r + \zeta v$, with cached released origin x_r and the first-step lowest transverse eigenvector v in its fixed sign convention. The original two retained coordinates and strain control stayed fixed. Here ζ is an internal length, not the external substrate coordinate t of Appendix G.2. Amplitudes $\pm 0.05/\pm 0.10$ Angstrom, both original quadratic predictions and a four-attempt cap were frozen before these labels; the center was reused. We reported the distinct finite-amplitude quantities

$$\kappa_E(h) = \frac{E(x_r + hv) + E(x_r - hv) - 2E(x_r)}{h^2}, \qquad \kappa_g(h) = \frac{v^T g(x_r + hv) - v^T g(x_r - hv)}{2h}.$$

They coincide for an exactly quadratic energy, but need not coincide at finite amplitude in an anharmonic profile. Neither difference is a total physical-error bound. These were training structures, not independent targets.

G.6. Local curvature, optimizer and finite-model definitions

For the zero-strain T-escape and preserving-switching sampled peaks, the local length-metric chart contains 36 translation-free internal coordinates: 33 atomic coordinates and three components of the open cell row. The prescribed substrate control is separate. Its length scale is $L = 5.12968067458423$ Angstrom; this chart is not the mass-weighted atomic projection used in the representation figures. Central differences of the physical energy gradient use steps 0.01 and 0.02 Angstrom in each of 37 internal-plus-control directions, with two signs per step. The cached center is reused: each center requires 148 new SCFs, 296 in total. Control probes use $\delta\epsilon = h/L$, not the reserved physical +0.5% condition. Raw directional responses are retained; spectra below use the symmetrized internal block only. Finite-step disagreement and symmetrization changes are diagnostics, not physical error bounds or a stationary-index certificate.

For a bounded conditional-relaxation comparison, the two negative eigendirections of the first-step internal matrix define fixed retained coordinates. The remaining 34 directions are released. Both optimizer arms start at the same cached center with the same retained coordinates, released-gradient norm target 0.03 eV/Angstrom, initial trust radius 0.05 Angstrom, internal-displacement bound 0.5 Angstrom and minimum atomic separation 1.6 Angstrom. Each arm has a conservative limit of twelve new SCF attempts, including unsuccessful calls; cache hits are free. One arm uses the default identity initialization. The other uses the measured Hessian restricted to a metric-orthonormal released basis, $U^T H U$, as a standard quasi-Newton initialization. Cholesky factorization must succeed without clipping eigenvalues. This algebraic check does not certify stability after relaxation. The conditional norm target is not the ordinary NEB residual threshold. Preparation and all trial calls are charged separately; a budget-exhausted arm supplies a censored cost, not a completed denominator for a speedup factor.

A separately registered branch trial resolves the shallow omitted internal direction explicitly. At each of five fixed Q pairs, two independent starts are displaced by $\zeta = \pm 0.10$ Angstrom from the original conditional origin along the same unit direction v_ζ. Both zeta and the other permitted released coordinates can move; the start sign is not a constraint. Safeguarded BFGS requires both $\|U^T g\| \leq 0.03$ eV/Angstrom and $|v_\zeta^T g| \leq 10^{-4}$ eV/Angstrom in the internal length chart. The additional test prevents an unresolved shallow direction from passing solely because its force is below the norm threshold. A positive numerical preconditioner combines the measured 33-direction restriction with a directional stiffness fitted to the four already seen signed gradient probes. It does not replace the original physical Hessian or certify release stability. Each start has the same twelve-attempt cap, geometry guards and electronic contract. The pre-label protocol and exact optimizer retain both starts and their costs; the bounded training outcomes are reported in Appendix F.1. No further start or radius adjustment followed.

We used all 81 energy/full-gradient observations from this trial, including rejected steps and budget-exhausted starts, for one finite model comparison. They were physical point observations, not 81 conditional minima. At each original curvature step separately, the measured 34 columns fixed the free block and its coupling to Q. The unmeasured symmetric Q/Q block supplied three fitted coefficients. The comparison family added seven terms, $\zeta^3, \zeta^4, Q_i\zeta^2, Q_iQ_j\zeta^2$ $(1 \leq i \leq j \leq 2)$, and the 99 terms $r_j\zeta^2, r_jQ_1\zeta^2, r_jQ_2\zeta^2$ over the remaining 33 coordinates. The harmonic and nonlinear families therefore fitted 3 and 109 coefficients, respectively, keeping the same native origin energy and gradient. Coordinates were normalized by 0.1 Angstrom; gradient equations were multiplied by 0.1 Angstrom before column-scaled least squares with relative SVD cutoff 10^{-10}. Both original steps and five leave-one-Q-family-out diagnostics were retained, without choosing a degree, weight or step from independent labels. The registered ansatz and exact fit preserve the raw negative direction; a positive bath at the source point does not certify stability throughout the fitted domain.

G.7. Literature boundaries for the mode and pathway comparisons

The geometric descriptors used here do not replace the symmetry-resolved mode analyses of earlier hafnia studies. Our Cmma and T reference controls are compared in their stated metrics and ranks. The

homogeneous paths do not compute a domain-wall mechanism. The public Ma–Liu v3 supplement provides examples of distinct oxygen and mixed Hf–O M-to-PO mappings; the present M candidate is not assigned to one by barrier-height agreement.

The supplementary Part 2 of Lee, Lee and Yu compares coupled-mode interpolation with restricted relaxation. Its Figure S3 uses a coupled interpolation parameter and cell energies, whereas Table S1 gives coefficients per formula unit. Those definitions differ from an initial/final-well response control and from the joint local q used here. Consequently, relaxation-induced profile lowering is not presented as a new method, and no literature mode coefficients are transferred into our energy model. (Lee et al., Supplementary Part 2 and Figure S3)